

**Media Engineering and Technology Faculty**  
**German University in Cairo**



# Design and Implementation of 6-DOF Robotic Arm

Bachelor Thesis

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| Author:          | Ahmad Mohamed Hisham |
| Supervisor:      | Dr. Eng. Alaa Khamis |
| Submission Date: | 1 June, 2008         |

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This is to certify that:

- (i) the thesis comprises only my original work towards the Bachelor Degree
- (ii) due acknowledgement has been made in the text to all other material used

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Ahmad Mohamed Hisham

1 June, 2008

# **Abstract**

This thesis is about designing and implementing a physical robotic arm that is capable of moving freely like human arms. This robotic arm can be used as an industrial robot. The thesis provides different technologies for designing the industrial robots. In addition to that a proposed design is given along with all the steps of constructing it from scratch and all the control units for it as well. Almost all of the topics that were used in the design were discussed clearly before they are applied.

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# Chapter 1

## Introduction

### 1.1 Motivation

Industry is the process of converting the raw materials to useful products that can be directly used by human. It is considered to be a major component in the developed countries. Each country has to dedicate some of its resources to be used in industry. Industry process uses many persons from different study fields. At first, we need persons who know the characteristics about the raw material and how it can be extracted from earth. Then another team is needed who can define accurately the requirements for the products to be generated by the industry. Also, some designers who can make a design for that tool from the available requirements, is needed. The design will include the procedures that should be taken to generate the product. After that, the role of the worker is to perform the procedures to create the product. Then the quality control team starts to check the quality of the products. Also, some workers may be enrolled in the packaging and transferring of the products. Taking a look at this long process, it is clear that there must be a more autonomous way to achieve the same process but quicker. But who is the team that is stressed by tasks that are achievable without using the human brain. It is the workers team. The worker has nothing to do except of obeying the procedures giving to him by the designer. In a same way, we can program the robot to do the same tasks that were given to the

worker. By that replacement, we got a worker robot that will never get tired, make mistakes, have salary and at the same time we can guarantee that all the products will be identical to each other. This thesis will introduce a simple design of a robotic arm that is reliable and can be programmed to do any required task.

## **1.2 Thesis objectives**

The objective of this thesis is to build a 6-DOF revolute arm that closely resembles the human arm. This robot can be used as a co-worker in the industrial process. In specific, we need to design a robot that is configurable, reliable and can be used in variety of environments. To prove that it is configurable (i.e. we can configure it to do any rule that we need); we have to choose a rule and make adjustments on it. Then allowing the robot to execute it. It should perform exactly the rule with our adjustments without any change or error. We can repeat that test for any other rule and also asserting that it works as expected. Extending this prove, we can test for reliability. To prove that it is reliable (i.e. obeys the rules without any change or error); we have to test it using a number of pre-defined set of rules and make sure that the output is identical to what we have expected. After passing these 2 tests, we can guarantee that the software part of the robot is both configurable and reliable.

As for the third requirement (i.e. can be used in variety of environments), it is very hard to test that it will work correctly in all environments. But if we have constraints on the environments, we can make a general test that it will work in these specific environments. For example, if we are going to use the robot instead of a human, then we can test that the robot arm can reach any point in space that the human can reach by his/her arm. Passing this test, then we can call that robot as “a general purpose robot”.

## **1.3 Thesis structure**

This thesis consists of six chapters including this one and five appendices. The second chapter introduces the field of Industrial robots. It highlights the currently available technologies used in designing the robots. Also some of tasks that can be done by the robots are mentioned. Narrowing down the types of the robots that we discussed in chapter two will lead us to chapter

three. In this chapter, we can see the currently available technologies used in designing a 6-DOF robotic arm. The proposed design is presented in chapter four and chapter five. Chapter four will discuss the integration of the system from hardware point of view and chapter five will discuss the same but from the software point of view. Finally, chapter six will conclude the thesis and suggests some future enhancements that can be done. Appendix A lists the datasheets of the used servo motors. Appendix B lists the budget summary of the whole design. Appendix C prints out the machinery report of the “TRUMPH” that is used in cutting of the aluminum sheet. However, Appendix D lists the area calculation of the aluminum sheet used and the schematic of each part. Finally, Appendix E is a tutorial for the board used in the development of the project, which is the easyPIC4.

# Chapter 2

## Industrial Robots

### 2.1 Introduction

Industrial robots are defined by ISO (International Organization for Standardization - [www.iso.org](http://www.iso.org)) as:

“Automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes”

In other words, they are general purpose, reprogrammable manipulators used in moving materials, or tools through a well-specified, variable programmed path for the performance of different tasks that are industrial related. These tasks can vary from a simple pick-and-place task and spray painting to a very accurate task like the welding of electrical components on printed circuit board and the construction of the micro-processors chips. This variation of tasks is the reason behind the calling them general purpose manipulators. They are also called reprogrammable. This means that their own software that controls their motion can be modified/replaced so that it will have different motions. This feature is extremely important, because using only one design we can distribute them in a way that each one has its own motion and from top point of view they are helping each other to perform more sophisticated tasks.

## 2.2 Basic Terminology

### 2.2.1 Degree of Freedom (DOF)

It is the ability of the robot to move in one axis. This corresponds to the existence of a joint in that axis. The type of the joint (translational or rotational) doesn't matter in counting the number of Degrees of Freedom. The human arm has 7 degrees of freedom; 3-DOFs in the shoulder, 1-DOF in the elbow and 3-DOFs in the Wrist. Any extra DOF than 6-DOFs is redundant (i.e. it doesn't provide a bigger working space). However, it provides greater flexibility in the movement (i.e. it can reach the same point by many ways, in order to avoid obstacles).

### 2.2.2 Joints

They exist to connect to rigid bodies to provide some movements. In most of these cases they represent Degrees of Freedom of the robot. As shown in Figure 2.1, there are many types of joints such as cylindrical joints (2-DOF), spherical joints (3-DOF), planar joints (2-DOF), screw joints (1-DOF), prismatic joints (1-DOF), and revolute joints (1-DOF). Prismatic and Revolute joints are most commonly used ones.

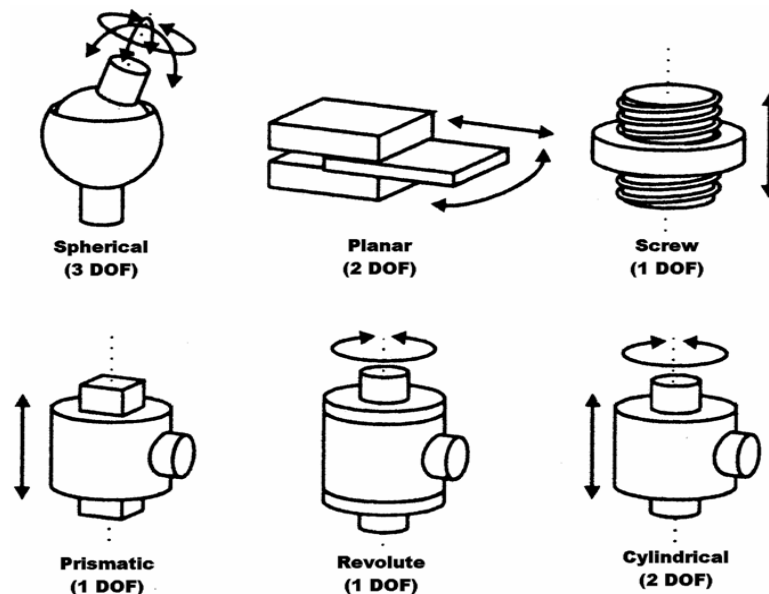


Figure 2.1: Robot Joints

### **2.2.3 Work Envelope or Working space**

The space in which the robot gripper can move with no limitations in the travel other than those imposed by the joints.

### **2.2.4 Accuracy**

It is the measurement of the distance between the desired target point in space and the actual point that the robot moves to. The lower is this distance, the higher is the accuracy.

### **2.2.5 End Effectors**

End effector is a tool at the end of a robot arm that is used to grasp or engage objects. Examples include, but not necessarily limited to, grippers, pushing hands, wrench tools, witch hands, welding torch, spray-coating gun, etc.

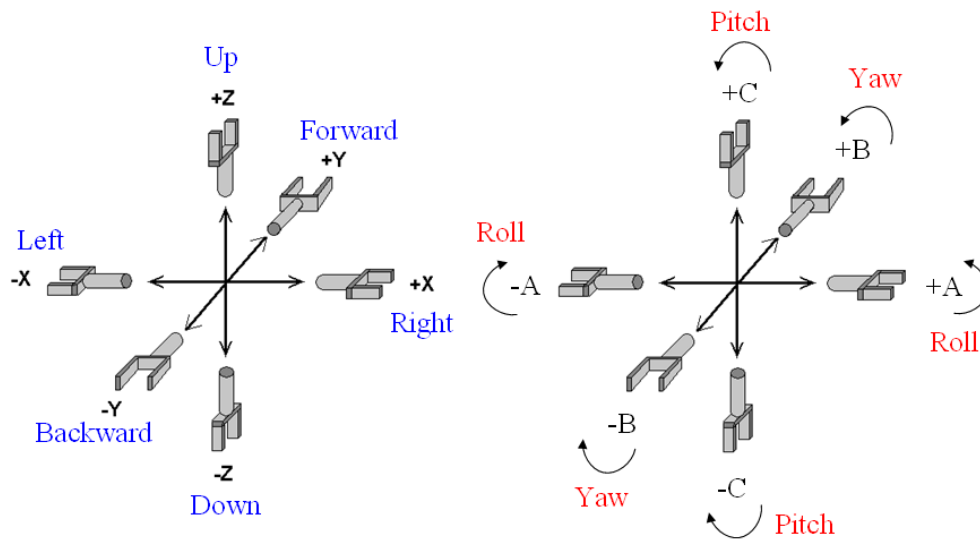
### **2.2.6 Working piece**

This is the object that needs to be manipulated by the robot. In most models, this object is fixed and the robot moves around it performing the industrial task needed. This is not the case in the Parallel Robot, where the object is directly attached to the robot and moves freely with it, while the manipulating tool is fixed in a certain position.

## **2.3 Geometric Classification of Robots**

Theoretically speaking, the robot must have at least 3 degrees of freedom in order to reach any point in the Cartesian space. They are usually called x, y and z coordinates. Another 3 degrees of freedom are needed in order to handle the orientation at a specific point. They are usually called Yaw, Pitch and Roll as shown in Figure 2.2.



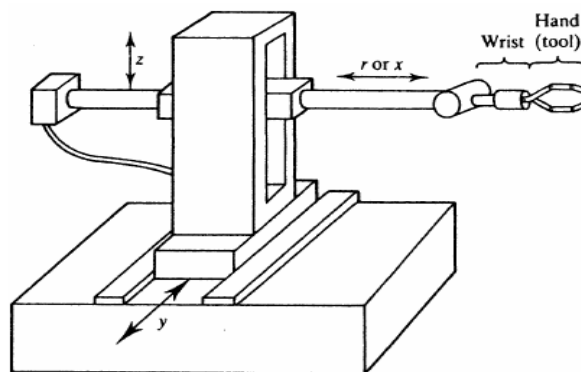


**Figure 2.2: Translational and Orientation Degrees of Freedom**

The robots are usually classified into 6 classes according to their geometry:

### 2.3.1 Cartesian (rectangular or gantry) robot

As shown in Figure 2.3, it has 3 translational joints. This means that it can only move in rectangular orthogonal directions. It has very high level of accuracy. It has rectangular working space. It is the easiest model. If we want to go to some point ( $P_x, P_y, P_z$ ) in space, then we just translate each joint by the amount specified in the corresponding axes.



**Figure 2.3: Cartesian Robot**

### 2.3.2 Cylindrical (post type) robot

It has 2 translational and 1 rotational joints as shown in Figure 2.4. It has less accuracy than the Cartesian robot, but more popular than it. It has cylindrical working space. It is relatively easy model. If we want to go to some point  $(P_x, P_y, P_z)$  in space, then we directly map the  $P_z$  value to the vertical translational joint. In order to get the values for the other 2 joints, we will have to consider the xy plane that the joints are involved in. Assuming that the horizontal translational is denoted by “R” and the rotational joint value is denoted by “ $\Theta$ ”. Then

$$R = \text{SQRT}(P_x^2 + P_y^2)$$

$$\Theta = \tan^{-1}(P_y/P_x)$$

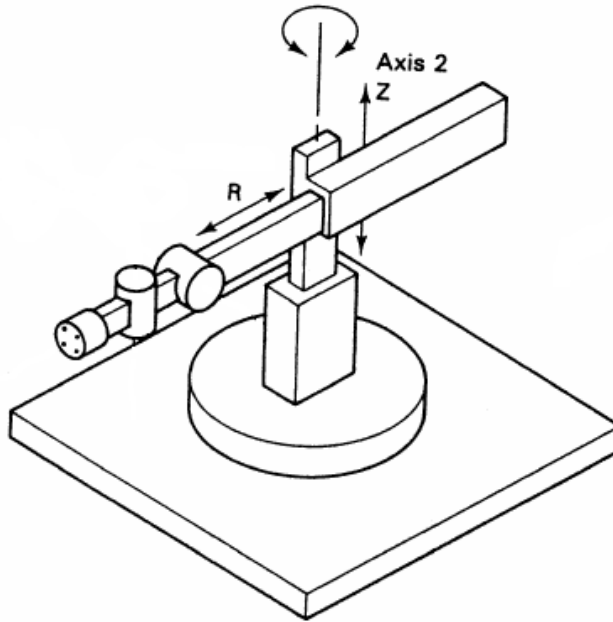


Figure 2.4: Cylindrical Robot

### 2.3.3 Spherical (polar type) robot

It has one translational and two rotational joints as shown in Figure 2.5. It has less accuracy than the cylindrical robot, but has more versatility (the ability to do more functions). It has a spherical working space. If we want to go to some point  $(P_x, P_y, P_z)$  in space, assuming that the

translational joint is denoted by “R”, the first rotational joint by “ $\Theta$ ” and the second rotational joint by “ $\Phi$ ”. We will have to calculate the values for all the 3 joints as follows:

$$R = \text{SQRT}(P_x^2 + P_y^2 + P_z^2)$$

$$\Theta = \tan^{-1}(P_y/P_x)$$

$$\Phi = \sin^{-1}(P_z/R)$$

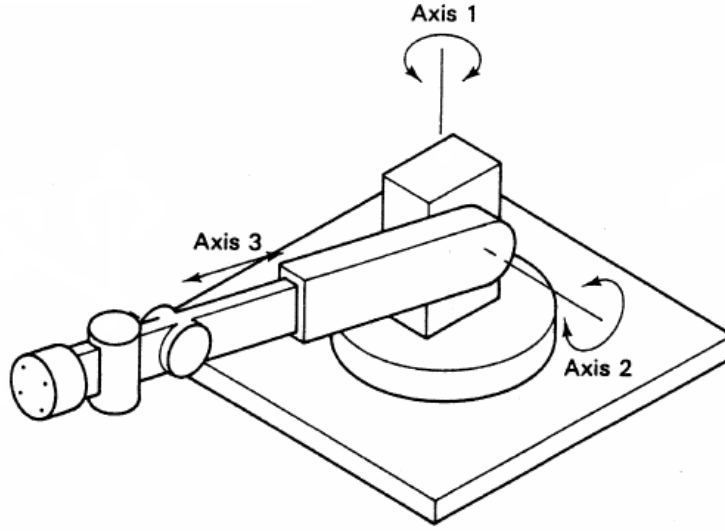


Figure 2.5: Spherical Robot

#### 2.3.4 Articulated Robot (Jointed Arm, revolute, or anthropomorphic robot)

This robot has all joints as revolute joints as depicted in Figure 2.6. It has the highest degree of versatility over all robots, and on the other hand, has the lowest degree of accuracy. It is used in environments where obstacles may exist. Its kinematics equations are not that obvious like the other models. It will be the main model in this thesis.

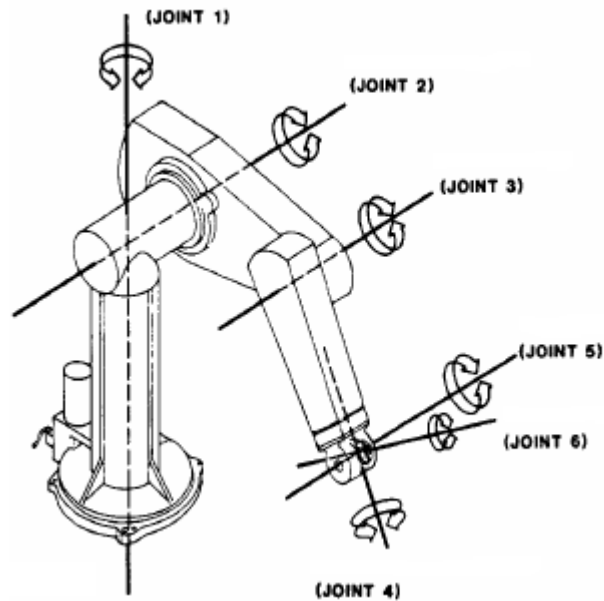
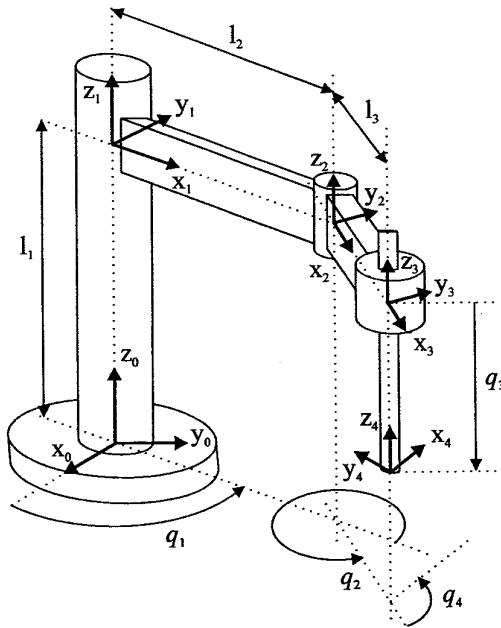


Figure 2.6: PUMA 560 Articulated Robot

### 2.3.5 SCARA ((Selective Compliance Assembly Robot Arm)) robot

SCARA was created in 1981 at the Yamanashi University in Japan. It is horizontally articulated robot as shown in Figure 2.7. It has all its joints of type rotational joints (or except one translational joint). All the joints revolute in plans that are parallel to each other. It is less versatile than the vertically articulate robot and hence more accurate than it. As is the case with the vertically articulate robots, the kinematics equations are not trivial.



**Figure 2.7: SCARA Robot**

### 2.3.6 Parallel robot

This is a very different geometry than all other models. This Robot is directly connected to the working piece while the tool itself is fixed in its place. That is different than the others where the tool is hold by the hand of the robot and the working piece is fixed in its place. It is stronger and more accurate than other models. But not all working pieces can be directly manipulated by it. For example we can not spray cars with fixed painter and moveable car. Also it can not avoid obstacles, on the other hand the Jointed Robot can. The kinematics equations are difficult. Figure 2.8 shows an example for a parallel robot.



**Figure 2.8: Parallel Robot (from: [en.wikipedia.org](http://en.wikipedia.org))**

## **2.4 Actuators**

It is the system that is responsible for driving all the axis of the robot. It is divided into 3 categories:

### **2.4.1 Electrical Actuation System**

It is composed of a source of electric power and an electric motor. The type of the electrical motor can be:

#### **2.4.1.1 Continuous Motor**

The continuous motor is used when a full rotations is required by the motor. It can be either a DC motor or a servo. The DC motor (Figure 2.9 a) has 2 terminals. Passing the current from one terminal to the other, makes the coil rotates around its axis. Reversing the current direction, will reverse the direction of rotation. On the other hand the servo (Figure 2.9 b) has 3 terminals. Two of them are for the VCC and GND. The third terminal is called the control signal. This signal is used by the servo-controller to detect the angle by which it will rotate. That is why the servo-motor is more accurate than the DC motor.



a) (Courtesy: [www.germes-online.com](http://www.germes-online.com))



b) (Courtesy: [www.bluemelon.org](http://www.bluemelon.org))

Figure 2.9: Continuous Motors

#### 2.4.1.2 Stepper Motor

The stepper motor (Figure 2.10) is similar to the DC motor, except that the internal coil is shaped as a gear. Also, the surrounding electromagnets are tooth shaped. When an external controller closes the circuit of one electromagnet, this electromagnet is called energized and will try to rotate the coil to adjust its gear with its tooth. This small rotation is called a step. The controller will iterate over all the electromagnets and energizes them in turn. Thus, it will result in a number of steps completing the full rotation.



Figure 2.10: Stepper Motor (from: [www.sherline.com](http://www.sherline.com))

### 2.4.2 Pneumatic Actuation System

It is a mechanical device that takes in any kind of gases and increases its pressure by decreasing its occupied volume. The range of pressure is typically between 10-50 bars.

### 2.4.3 Hydraulic Actuation System

They are similar to Pneumatic Actuation System but use compressed oil at 50-100 bars and sometimes up to 300 bars instead of compressed air.

### 2.4.4 Comparison

Table 2.1 summarizes the advantages and disadvantages of each type of the three actuation systems.

**Table 2.1: Actuation Systems**

|                      | Advantages                                                                                                                       | Disadvantages                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electrical Actuation | <ul style="list-style-type: none"><li>• Fastest and the most reliable</li><li>• Inexpensive motors</li><li>• Not noise</li></ul> | <ul style="list-style-type: none"><li>• Low torque (because of energy dissipation)</li><li>• Brakes are required to hold the position</li><li>• Can not be used in explosive environment</li></ul> |
| Pneumatic Actuation  | <ul style="list-style-type: none"><li>• Fast</li><li>• Inexpensive</li></ul>                                                     | <ul style="list-style-type: none"><li>• Noise</li><li>• Leakage of air (can be of concern)</li><li>• Difficulties with the speed control</li></ul>                                                 |
| Hydraulic Actuation  | <ul style="list-style-type: none"><li>• Can have very large payload</li></ul>                                                    | <ul style="list-style-type: none"><li>• Leaks eventually</li><li>• Noise</li></ul>                                                                                                                 |



|  |  |                                                                                                      |
|--|--|------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"> <li>Needs additional equipment (motor, pump, tank ...)</li> </ul> |
|--|--|------------------------------------------------------------------------------------------------------|

## 2.5 Sensors

Sensors are used by robots to figure out the surrounding environment. Technically speaking, the sensor is just a device that converts one type of energy to another (usually electrical). This transformation is done by using a transducer that can sense the original type of energy and a circuit to convert that sense. The electrical output of the sensor may be used by the controller to make decisions that depends on the status of that sensor. Also, this output is used in larger systems in perceiving the surrounding environment. Perceiving is the process of collecting all the necessary data from the sensors and interpreting them to form an understanding of the environment. The most important sensors types that measure physical variables related to industry are:

### 2.5.1 Sensor Classification

Sensors are usually classified into three categories:

- **Inner-state sensors**

Those are the sensors that measures internal variables of the robot. Examples are optical encoder and battery status.

- **Outer-state sensors**

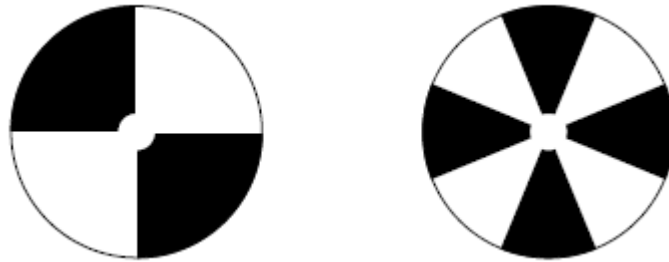
Those are the sensors that measures external variables of the environment. Examples are Ultrasonic, infrared and vision.

- **Surface sensors**

Those are the sensors that are on the surface of body of the robot. Examples are Tactile.

### 2.5.2 Inner-State Sensors

Optical encoder is used as inner-state sensors to measure the robot position and speed. This is done by placing a circular disk on the shaft of the motor as shown in Figure 2.13. This disk is divided into a number of equal regions of black and white. Or a number of holes are drilled into the disk. Infrared sensors will be placed on both sides of the disk. It will count either the number of black/white regions or the holes. The generated signal will be sent to the controller to determine the current speed/position of the motor. This process is called “Optical Shaft Encoding”.



**Figure 2.11: Optical Shaft Encoder Disks**

Precautions have to be taken into consideration when using the infrared sensors, because they may be influenced by the surrounding visible light.

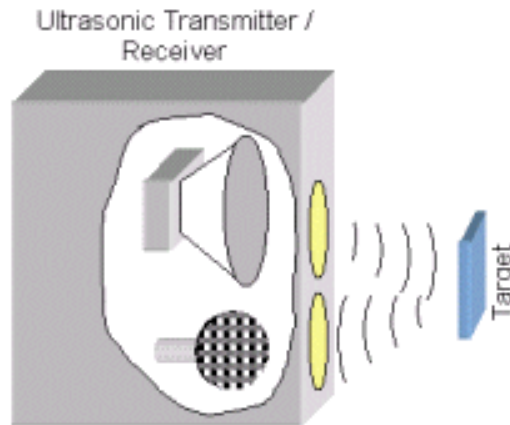
Another example for inner-state sensors is the motor current sensor. This sensor is used to measure the amount of current that is being drawn by the motor. It is used to detect the amount of load that is being exerted by the motor. It will be used by the controller to handle the stalling case of the motor. When the motor stalls, it will draw a large amount of current in order to overcome the stalling and continue moving. This large amount of current may burn out its controller resulting in a damaged motor.

### 2.5.3 Outer-state Sensors

As an example for outer-state sensors is the ultrasonic range finder. The main aim of the Ultrasonic sensors is to measure the distance between itself and another object. It works by emitting a sound wave of very high frequency that the human ear can not hear. Then it listens for

its echo. As shown in Figure 2.12, the sound wave will propagate in air, and when it collides with the object, it will be reflected back. Since the velocity of the wave is known, and we can measure the time interval between the emitting and receiving, we can directly substitute in the equation to measure the distance:

$$\text{Distance} = \text{Velocity} * \text{Time}/2 \quad [\text{round-trip}]$$



**Figure 2.12: Ultrasonic Sensor (from: [www.euclidres.com](http://www.euclidres.com))**

The main application for the Ultrasonic sensors is the collision avoidance robots. Those are robots that can freely move in the environment. They use the ultrasonic sensors to know the places of objects so that they do not collide with them.

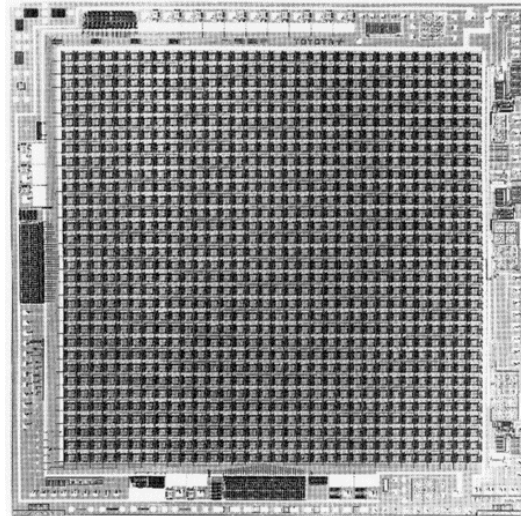
This sensor was inspired from bats in nature. Bats are totally blinded. They use a similar natural phenomenon to determine the existence of the objects and therefore do not collide with them.

Vision is considered an enhancement for the robot if it can see its working piece. Using cameras, the controller will be able to visualize the surrounding environment. Also, it may be used by the controller to take precise decisions in performing the industrial operation. The disadvantage of using vision camera is that the visualizing software is relatively large and usually doesn't fit on small controllers.

#### **2.5.4 Surface State Sensors**

Tactile sensor is an example for surface state sensors. Tactile sensors represent the touch sense of humans. Usually, they consist of a touch surface that composed of a matrix of pressure pins,

along with some electric components as shown in Figure 2.13. They are used to sense the presence of objects, detecting the part shape and in some advanced sensors; they can calculate the force and moments magnitude.



**Figure 2.13: Tactile Sensor Array – Artificial Skin (from: [www.wtec.org](http://www.wtec.org))**

## **2.6 Applications of Industrial Robots**

### **2.6.1 Material Transfer**

This is the most important task of the robot arm. This is also called pick-and-place task. The main aim of the robot is to hold some material from a specific place and move it to another specific place. In order to accomplish this task, we need to control the robot to teach him the two places and also how to grasp the material using the end-effectors without damaging it.

### **2.6.2 Welding**

The robot may be used to weld some materials to each other. There are two types of welding, spot welding (Figure 2.14) and arc welding (Figure 2.15). The spot welding process, the robot has to know the exact point at which the welding is needed. It uses high electric current that passes through the 2 sheets in the welding spot causing a fusion. On the other hand, the arc welding is a continuous process. The robot holds an electrode and moves along a specific path

over the 2 sheets to be welded. When the electrode touches the metal sheets, a very high electric current passes that causes the welding. The arc welding is usually harmful for humans because it produces ultraviolet rays. So that is a perfect task to be done by robots.

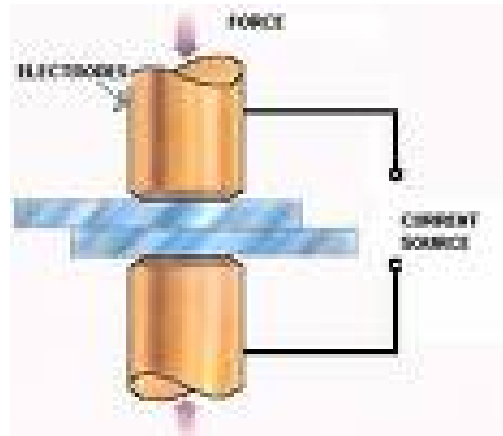


Figure 2.14: Spot Welding (from: [www.robot-welding.com](http://www.robot-welding.com))

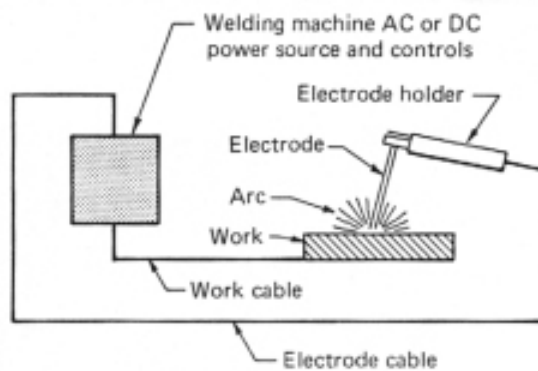


Figure 2.15: Arc Welding (from: [www.robot-welding.com](http://www.robot-welding.com))

### 2.6.3 Spray Coating

Robots are used to change the color of the product before delivering it to the customers. The robot usually holds a spray gun and knows how to open it. A specified area of painting is defined for the robot. The robot then moves the spray gun within the painting area to paint it. As Arc welding, the spray coating is unsafe for humans due to the fumes and the mist in air. So spray coating is a typical task to be done by robots rather than by humans.

#### **2.6.4 Product Assembly**

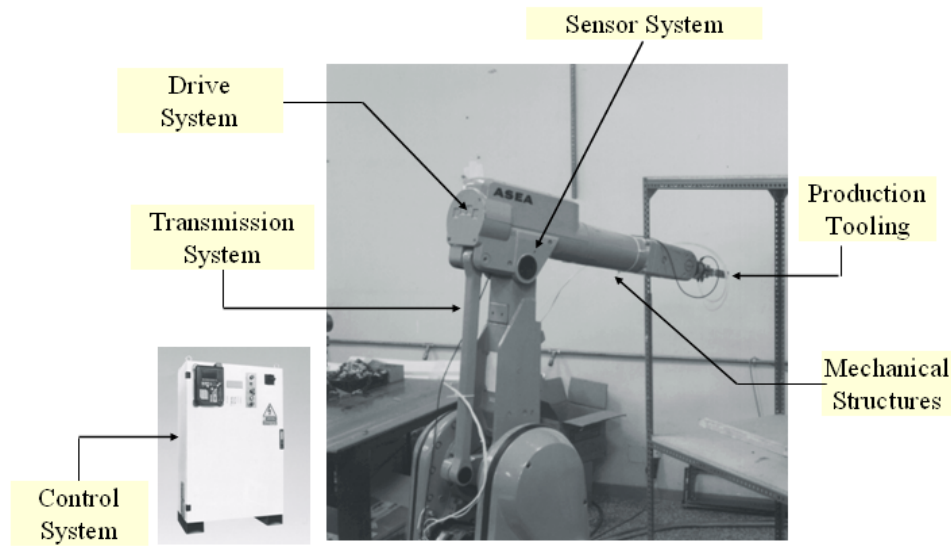
The robot may also be used to assemble a group of sub-products to form the finished product. To do this task, all the sub-products must be in the working envelop of the robot and their places are well-defined for the robot. Also the assembly procedure must be programmed in the robot. The robot will then execute these procedures to assemble these sub-products to each other. There might be some screws in the sub-products. So the robot has to know how to handle them. One way might be to change the end-effector to hold a screwdriver and use it in fastening the screws correctly. Also some welding might be used to assemble the sub-products.

## **Chapter 3**

# **Morphology of Robotic Arms**

### **3.1 Introduction**

Human Arms are 7-DOF. However, the 6-DOF Arm is a very good replacement for the 7-DOF Human Arms. We are allowed to this replacement because of the fact that ensures that any position in space can be reached by its 3 coordinates (x, y and z). In addition to that, the orientation of the end-effector at the target position can be obtained by having another 3 values (yaw, pitch and roll). As a result, we need 6-DOF to reach any point in space with any orientation of the end-effector. However, the extra DOF of the Human Arm serves as a flexibility advantage for the arm. This chapter will introduce the familiar technologies in the field of designing the 6-DOF Robotic Arm. This chapter introduces the morphology of robotic arms addressing the 5 subsystems that form a robotic arm as illustrated in Figure 3.1.

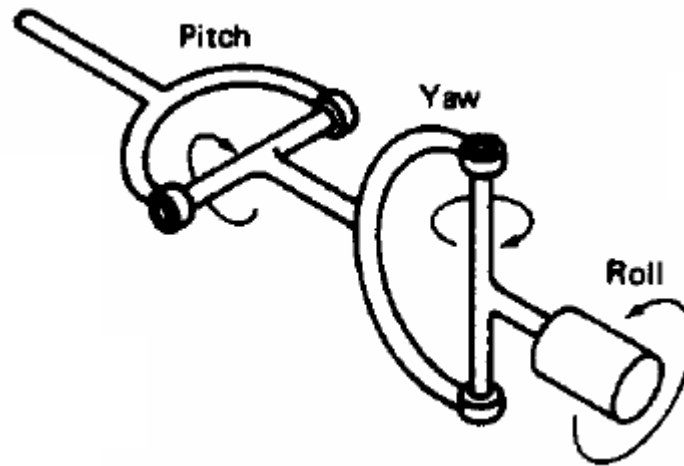


**Figure 3.1: Morphology of a Robotic Arm**

## 3.2 Mechanical System

This is the main system that describes the mechanism of the robotic arm. Since we have 6-DOF robotic arm, therefore we will need at least 6 different joints. Joints (as described earlier in section 2.2.2) are the links that connects two or more rigid bodies together to form additional Degrees of Freedom. Joints are available in two types, translational and rotational joints. The combination of different joints that are used in modeling the robotic arm is used to determine the geometric class that it belongs to. The Wrist is the joint of greatest interest in the design. It is used to connect the end-effector to the rest of the arm. The aim of all other joints is to deliver the end-effector to a specific point in the working envelop. In addition to that, the main aim of the wrist joint is to orient the end-effector in the target position. To accomplish this aim, the wrist must have 3-DOF to represent any orientation of the end-effector. These 3-DOF are called Roll, Pitch and Yaw as shown in Figure 3.2.





**Figure 3.2: Robotic Wrist**

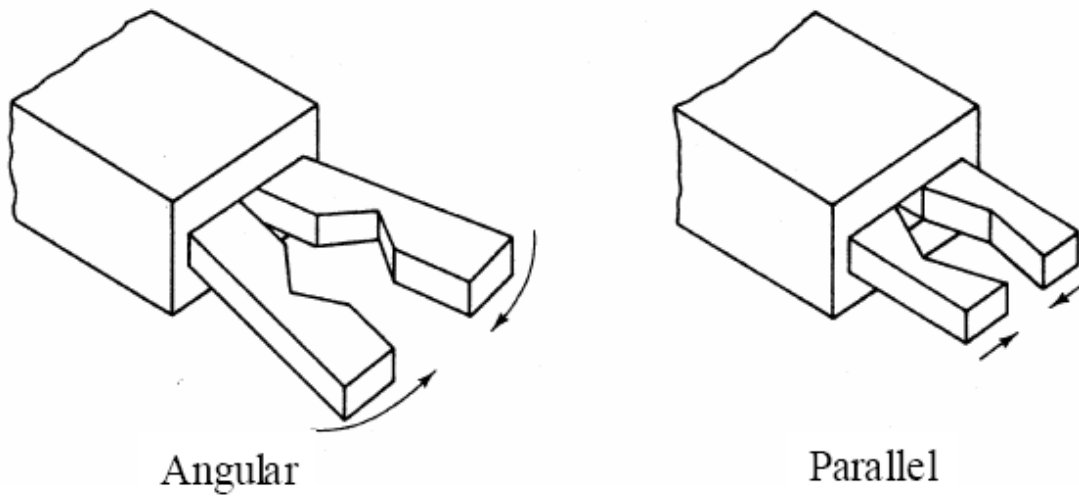
This design belongs to Jointed Arm Geometric class. Therefore, all the Joints are of type rotational. The Joints are distributed as follows:

- 3 Joints for the base, shoulder and elbow
- 2 Joints for the wrist (roll and pitch only)
- 1 Joint for the fingers

This distribution sums up to get 6 Joints. However, 8 motors are commonly used. This is because of both the shoulder and wrist (pitch movement) use 2 corresponding motors to increase the resultant torque.

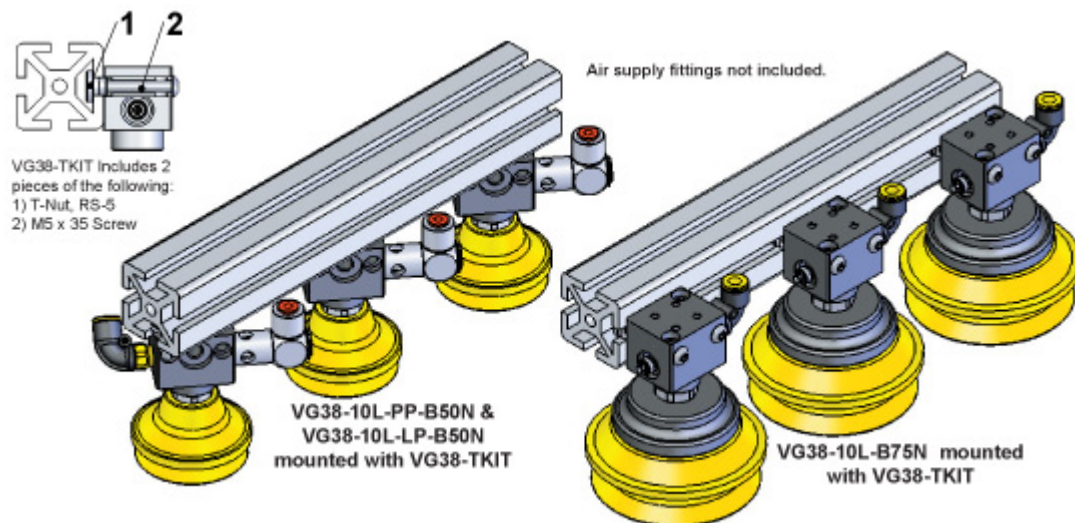
### **3.3 Production Tooling System**

This is also called the end-effector and it is the system that is being attached to the wrist and is responsible to do the production process. Examples of well-known end-effectors are Grippers, Welder, Spray coater and Wrench Tools. The grippers must be able to grasp, hold and release the working piece. In addition to these tasks, it might be able to sense the working piece. This sense is helpful to detect whether the working piece is correctly grasped or not. This will prevent the damage of fragile pieces. Standard Grippers are usually classified into 2 classes according to the opening/closing motions, angular and parallel grippers as illustrated in Figure 3.3.



**Figure 3.3: Angular and Parallel Grippers**

Another type of Grippers is the Vacuum Gripper (Figure 3.4). Vacuum Gripper usually consists of a vacuum cup where it is fully aligned with the surface to be grasped. Then the enclosed air between the cup and the surface is fully pulled out resulting in a very low pressure. The result is that the surface is adhered to the cup. To release the surface, we have to open the cup, thus resulting in normalizing the pressure.



**Figure 3.4: Vacuum Cups (from: [www.edcousa.net](http://www.edcousa.net))**

The ability to change the end-effector during the execution of a specific task using a tool-changer is of great advantage to the robot.

The proposed design in this project uses the standard angular gripper. The gripper is composed of 3 fingers to grasp the working piece. The contacting area in each finger has small touch sensors that will detect the grasping of the piece.

### **3.4 Drive System**

It is the system that is responsible to generate the movement for the robotic arm. As we discussed earlier in section 2.4, they are divided into 3 categories: Electrical, Pneumatic and Hydraulic Actuators.

The proposed design in this project uses the Electrical Continuous Servo Motors as the main drive system. Eight servos have been used to represent the 6-DOF of the robotic arm (see section 3.2). The base servo is the GWS125 Sail winch servo which is capable of continuous rotation (360°). The shoulder and elbow use the Hitec HS-5645MG servos which are Digital Ultra Torque servos. The wrist and the fingers use the Futaba S3003 standard servo. Appendix A provides more information on the used servos.

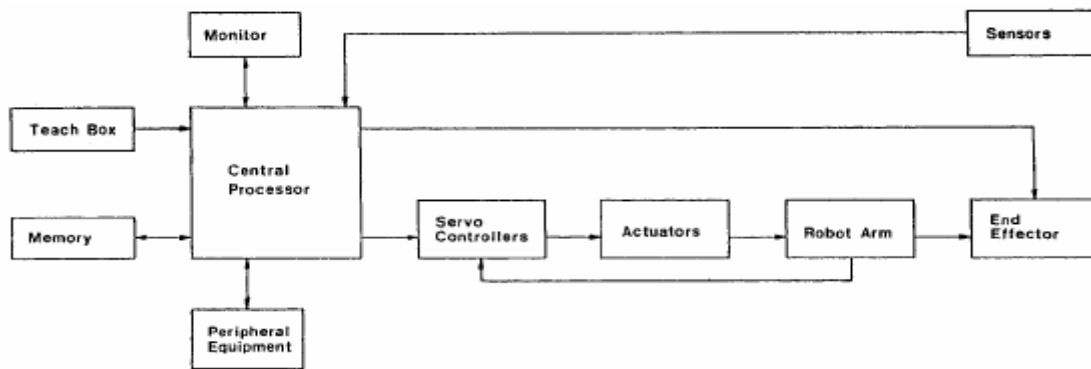
### **3.5 Control System**

It represents the brain for the robot. It is the part responsible for making decisions for the motors. It consists of a built-in firmware that is robot specific. The firmware consists of a logical sequence of commands, usually written in C or assembly. This firmware is executed on a microprocessor, or in case of large firmware (called operating system), it is executed on a computer. The controller is connected to some input devices (sensors), some output devices (motors) and to memory. Taking input from the sensors and manipulating it by the firmware commands using the memory, helps the controller to take decisions to control the motors.

There are two main categories for controllers, open-loop and closed-loop. The open loop controllers are those controllers that produce signals to motors, and don't really matter whether the motor did reach the target position or not. It depends on the controller of the motor itself. On

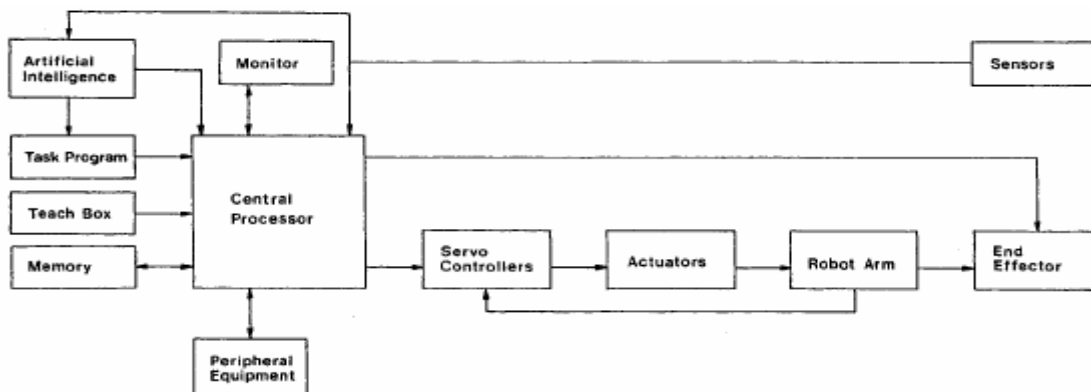
the other hand, the closed-loop controllers send the signal to the motor and then start monitoring the motor. This is called a feedback. Also, the controller may produce some helping signals to the motor, if it detects that the motor will not be able to perform the desired task.

Another Classification for the controllers is according to its intelligence level. In that way, the traditional controllers (Figure 3.5) use the built-in firmware along with the available values from the sensors and then produce a command for the motors.



**Figure 3.5: Traditional Controller**

On the other hand, the controller may use an artificial intelligence unit (Figure 3.6). As the traditional controller uses the built-in firmware along with the available values from the sensors, the intelligent controller uses them and produces the command but after asking the artificial intelligence unit.



**Figure 3.6: Intelligent Controller**

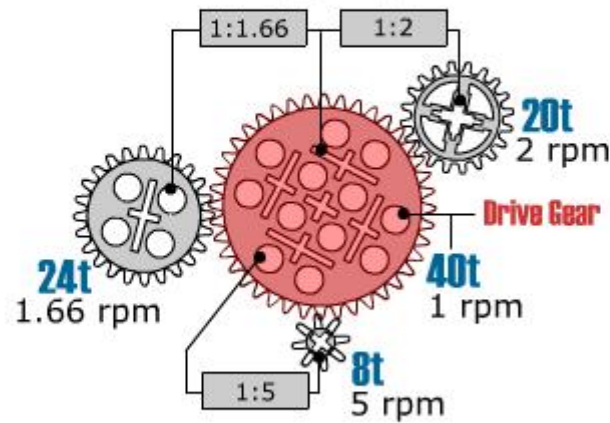
The proposed design in this thesis uses the pic16f877 as the main controller unit. The micro-controller is connected to a PC through the serial port RS232. This is considered as the main input stream. Also, the sensors in the contact area of fingers are connected to the micro-controller. The output of the micro-controller is in the form of variable width pulses which is used to control the position of the 8 attached servos. More details about the implemented control system can be found in chapter 5.

### **3.6 Transmission System**

This is the system that is responsible for transferring the motion from the drive system to the rigid body it is moving. Using Transmission system allows that the position of the drive system to be distant from the body it is moving. Usually the drive system is located in the base and using the transmission system, all the motion is transferred to the joints. This results in a smaller weight of the body, so a higher power is achieved. The most commonly used transmission system is the gears. Gears can be easily connected to each other to produce a system of adjustable torque to speed ratio. Belts and Linkages are other examples of transmission systems.

The proposed design didn't use any transmission systems except for the gear boxes that are already built inside the servos (Figure 3.7). The servo motors are directly attached to the joints they are moving.

## Increasing Speed



To **increase speed** and **reduce torque** use a large drive gear coupled to a smaller driven gear. In this example we use a 40 tooth drive gear to turn smaller gears to get the illustrated gear ratios.

So the 8 tooth gear will turn 5 complete rotations for the every rotation of the 40 tooth gear. The motor has been **geared up**.

Figure 3.7: Transmission system (from: myweb.cwpost.liu.edu)

### 3.7 Sensor System

This is the system used by the microcontroller to sense the surrounding environment. It may be used to test for the quality of the product. As discussed earlier (see section 2.5), the sensor system is usually classified into inner-state, outer-state and surface sensors.

The developed 6-DOF robotic arm in this thesis uses small touch sensors as surface state sensors to detect the grasping state. Those small touch sensors are located in the contact area of the fingers and are used to sense that the working piece is grasped well.

# **Chapter 4**

## **System Integration – Hardware**

### **4.1 Introduction**

The mechanical structure of this 6-DOF robotic arm was inspired from chapter 5 of the “Build Your Own HUMANOID ROBOTS” book with some modifications. This chapter will introduce the mechanical and production tooling systems that were implemented in this design. Then it introduces the electrical circuit of the controller.

### **4.2 Mechanical System**

This robotic arm is made of aluminum sheet that is 1m long, 50cm width and 1mm thickness. The robot base is rotating on a wooden box of dimensions 20cm width, 20 cm height and 11.5cm height as shown in Figure 4.1. The wooden box will contain the controlling circuit that will control the 8 servos used in the arm.

For more information on the used servos, refer to Appendix A.

For more information on the exact dimensions of the body pieces, refer to Appendix D. The report generated by the “TRUMPH” machine for cutting these pieces is in Appendix C.



**Figure 4.1: Wooden Box**

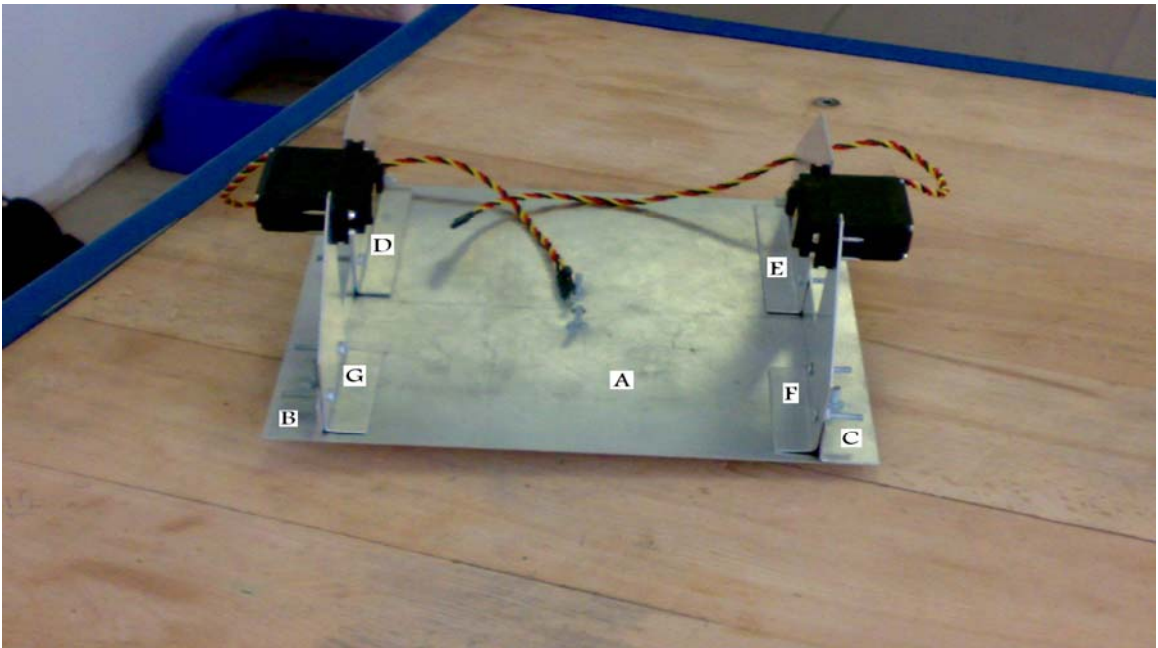
#### **4.2.1 Construction of the rotating base**

The rotating base is powered by a single “GWS S125 SAIL WINCH SERVO” which is capable of rotating in a continuous mode (i.e.  $360^\circ$ ). This servo has a white circular horn that is mounted in the center of piece A. The servo itself is attached to the top surface of the wooden box as shown in Figure 4.1. A Bearing is located between the rotating base and the top surface of the wooden box to support the robot from tilting.

The rotating base (Piece A) is constructed using a 20cm length and 20cm width. On the 2 sides of this base, Pieces B and C are located. They are used to hold Pieces D, E, F and G in a vertical position. On each side, there is a HITEC HS-5645MG servo that is mounted to the vertical pieces. These servos are used to rotate the shoulder. They are digital, re-programmable and produce high torque. The gears are made of metals, thus can hold weights without worrying about the destruction of the gears.

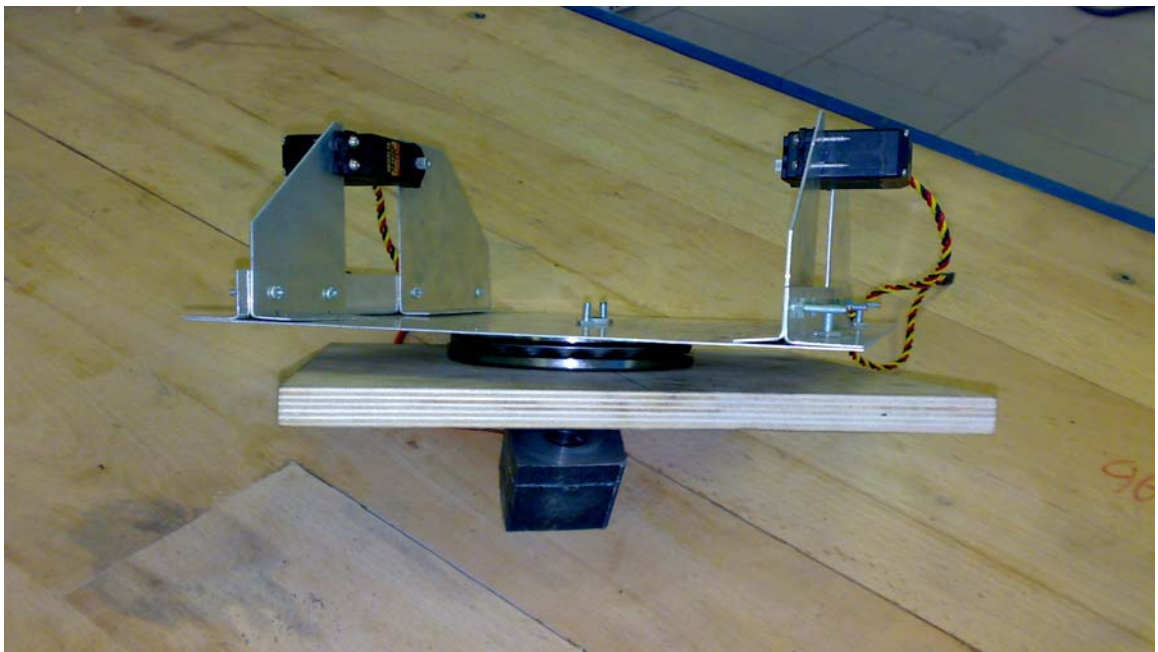


The assembled rotating base is shown in Figure 4.2.



**Figure 4.2: Rotating Base**

The rotating base with the top surface of the wooden box and the bearing is shown in Figure 4.3.

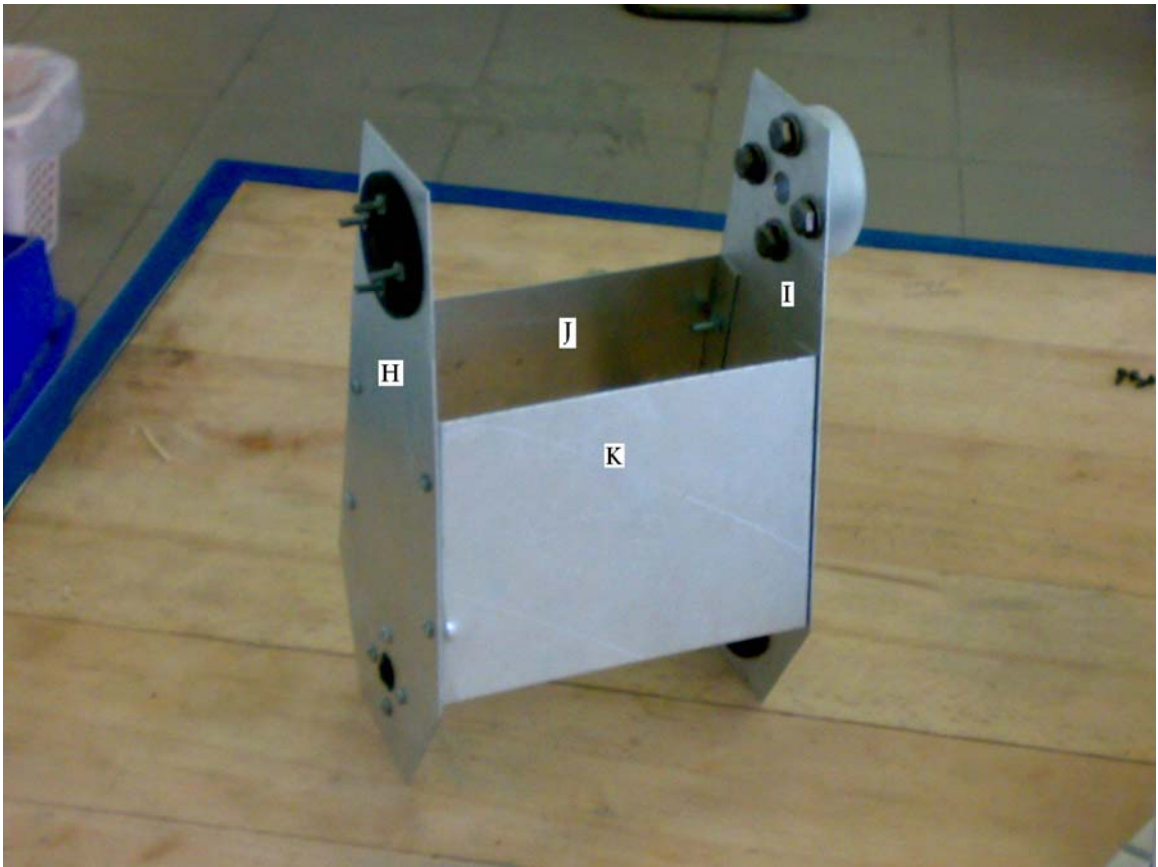


**Figure 4.3: Assembled Base**

#### 4.2.2 Construction of the shoulder

The shoulder is constructed from four pieces (Pieces H, I, J and K). Two horns are fixed in the 2 Pieces H and I. They are used to connect the shoulder to the motors attached to the vertical pieces in the rotating base. There is a bearing attached to the top of Piece I. This is used to mount the elbow to the shoulder and allowing the elbow to rotate easily without friction. On the other side (Piece H), a horn is fixed. This horn is used to attach the servo motor in the elbow. Pieces J and K are used to connect the two sides of the shoulder together.

The assembled shoulder is shown in Figure 4.4.

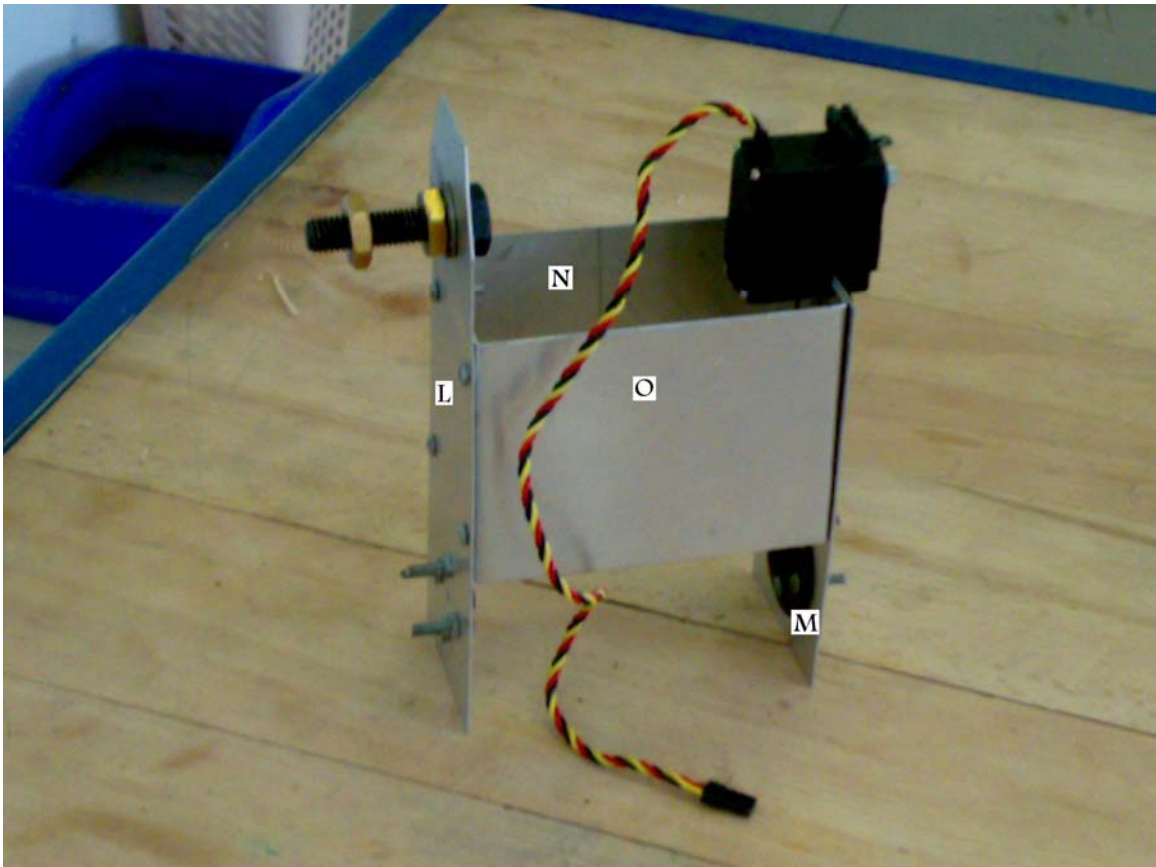


**Figure 4.4: Shoulder**

### 4.2.3 Construction of the elbow

The elbow is constructed from four pieces (Pieces L, M, N and O). Piece L is made longer than Piece M to allow it to be attached to the bearing on Piece I. On the other side, Piece M has an attached HITEC HS-5645MG servo that is responsible for the rotation of the elbow. It will be mounted to the horn fixed in Piece H of the shoulder. On the other end of Pieces L and M, two horns are fixed. They will be used to attach the two servos of the wrist. Pieces N and O are used to connect the two sides of the elbow together.

The assembled elbow is shown in Figure 4.5.



**Figure 4.5: Elbow**

#### 4.2.4 Construction of the wrist

The wrist is composed of 3 Futaba S3003 servos and 4 Pieces (Pieces P, Q, R and S). This is a standard analog servo that has a good balance between the speed and the produced torque. The two opposite servos are used to rotate the wrist providing the “pitch” orientation. The remaining servo is used to hold the hand and rotate it providing the “roll” orientation. The two pieces P and Q are used to attach the 3 servos together. The other two pieces R and S act as a cover for the wrist. Despite of the presence of the 3 servos, this wrist is a light component that offers 2 types of orientations.

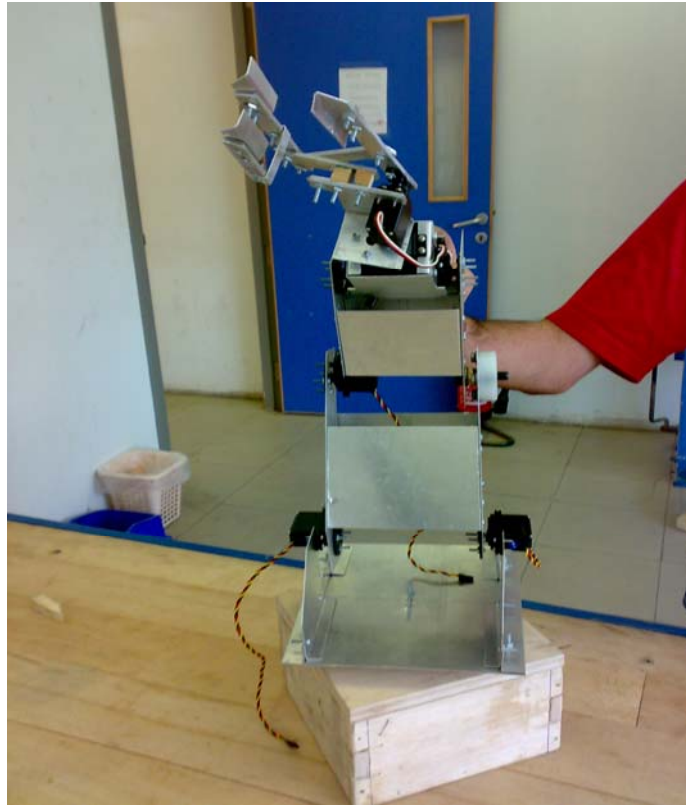
The assembled wrist is shown in Figure 4.6.



**Figure 4.6: Wrist**

The whole robotic arm after being assembled is shown in Figure 4.7.





**Figure 4.7: The whole robotic arm**

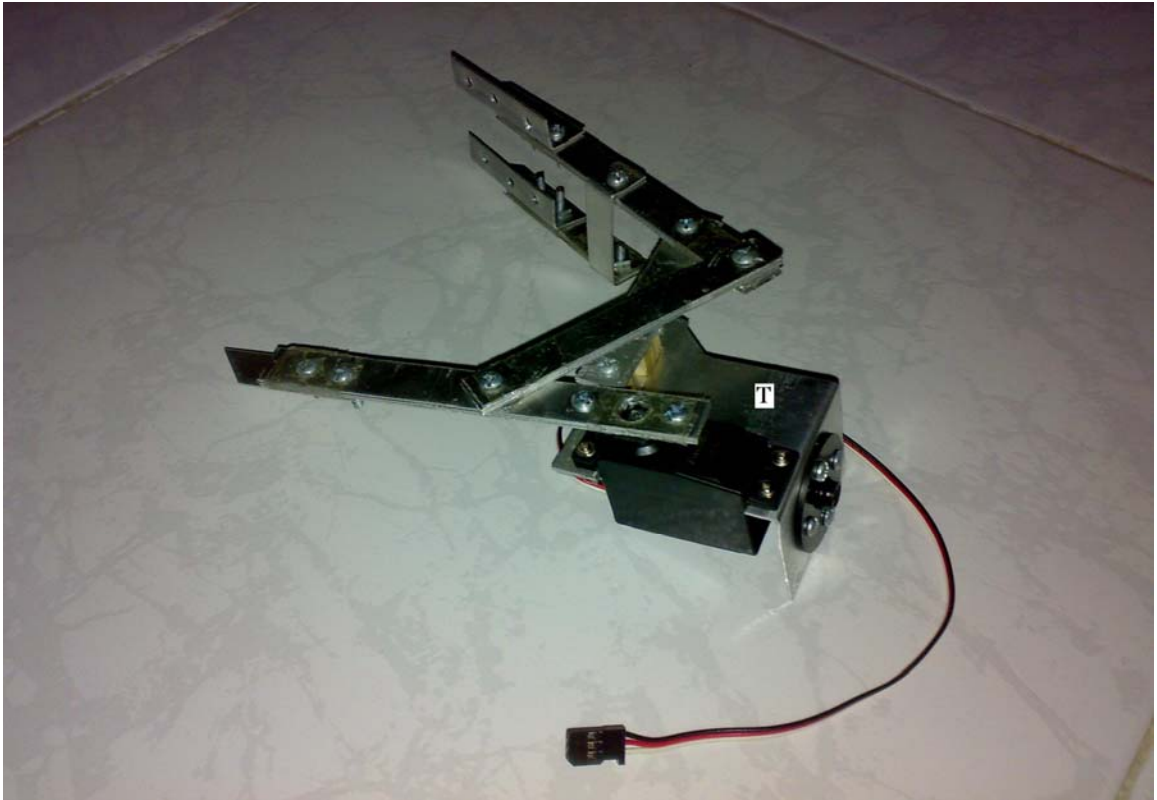
## **4.3 Production Tooling System**

This design has a fixed production tooling that is the gripper tool. The gripper will allow the arm to grasp and release objects.

### **4.3.1 Construction of the hand**

This design has 3 fingers. One of the fingers is directly attached to the servo motor. The servo motor that is used is the Futaba S3003 analog servo. The other two servos fingers are controlled via a direct mechanical linkage. The fingers are straight and don't offer the opportunity to be bended. The hand is constructed using 11 Piece. The Piece T is the main piece that holds the 2 fingers and the servo with its attached third finger. Also, there is a horn that is fixed on the back side of Piece T. This horn will be attached to the servo motor in the wrist that is responsible for the "roll" orientation.

The assembled hand is shown in Figure 4.8.



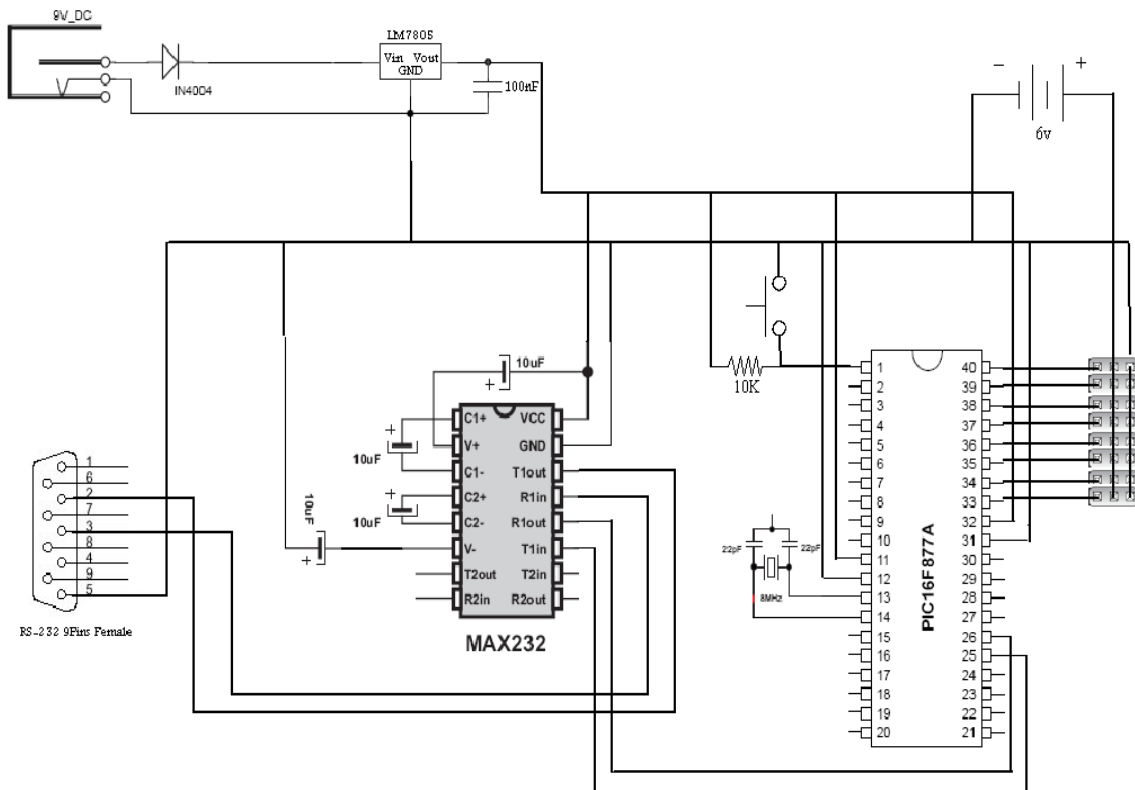
**Figure 4.8: The Hand**

## **4.4 Controller Circuit**

There is a main controller circuit where the RS-232 communication is implemented together with the pulses generator for the servos. Besides that controller, there is another controller that is used to act as an artificial skin of the fingers. The main function of that controller is to determine the level of grasping of the object in hand. Its output is then transferred to the main controller to be outputted on the GUI application.

### **4.4.1 Main Controller Circuit**

The schematic of the controller is shown in Figure 4.9:



**Figure 4.9: Main Controller**

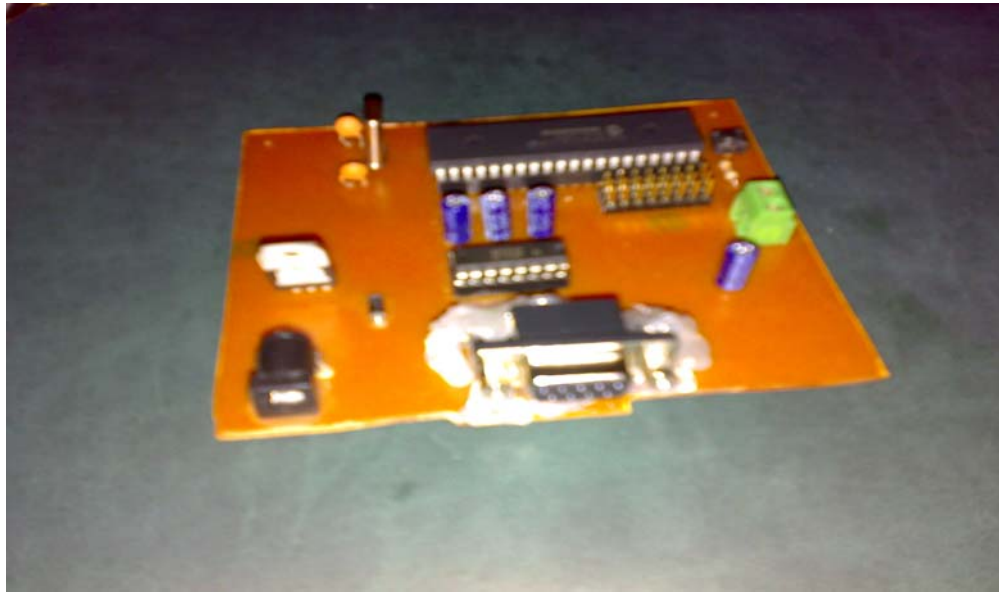
The controller that was used is the Microchip PIC 16F877 microcontroller. It uses a clock of 8 MHz connected to pins 13 and 14. It is important to use 2 different electrical sources for the PIC and the servos so that any generated noise from the servos doesn't interfere with the PIC. The electric source that is used is 9V DC adapter. It passes on a diode IN4004 to avoid the wrong polarity of the adapter (just in case). After passing through the diode, it becomes regulated to be 5V by the LM7805 voltage regulator. A capacitor of 100nF is used to stabilize the regulator. The regulated 5V is then connected to the PIC through pins 11 and 32. An important notice is that the whole circuit should have only one ground node. That means, the 9V adapter, the regulator, the RS-232 serial port, the MAX232, the PIC, the batteries and the servos themselves should be connected together. The ground node is connected to the PIC through pins 12 and 31. A hardware reset circuit is connected to pin 1 of the PIC. This is used to reset the PIC and starts its code from the beginning. Its idea is that the pin is always connected to the logical state "1". When the push-button is pressed, the current will flow through the push-button to the ground

leaving the pin 1 with a pure logical state “0” (which means hardware reset). The 8 pins of PORTB of the PIC are connected to the 8 control signals of the 8 servos. That is how the signal that will be generated inside the PIC is transferred to the servos. The electric source of the servos is taken to be a battery of 6V. Pins 28, 29 and 30 of the PIC (RD5, RD6 and RD7) are used to get the results from the grasping controller (section 4.5.1).

The serial port of a PC uses the RS-232 communication protocol. So the circuit has a RS-232 9Pin female connector. The two connectors will be connected using a serial cable that is male in one end (to be connected in the circuit) and a female in the other end (to be connected in the PC). The RS-232 protocol will be discussed later. For now, the MAX232 is used to convert the signals used by the RS-232 to CMOS logical levels that the PIC can manipulate. It has 2 receivers and 2 transmitters. Only 1 receiver and 1 transmitter are used. The Tx channel of the PIC (pin 25 – RC6) is used to transmit data out of the PIC. It is connected to the T1in of the MAX232. Its output pin T1out is connected to the Rx pin of the serial port (pin 2) to be transmitted to the PC. Reversing the trip, the Tx pin of the serial port (pin 3) is used to transmit data out of the PC. It is connected to the R1in of the MAX232. Its output pin R1out is connected to the Rx pin of the PIC (pin 26 – RC7). Pins 1, 4 and 6 of the serial port are connected together to allow the PC to read the data sent by the PIC at any time. Pins 7 and 8 of the serial port are connected together to activate the hardware flow control. Pin 5 is grounded while Pin 9 is unconnected (Ring indicator for modems).

The controller after being printed on a PCB is shown in Figure 4.10.





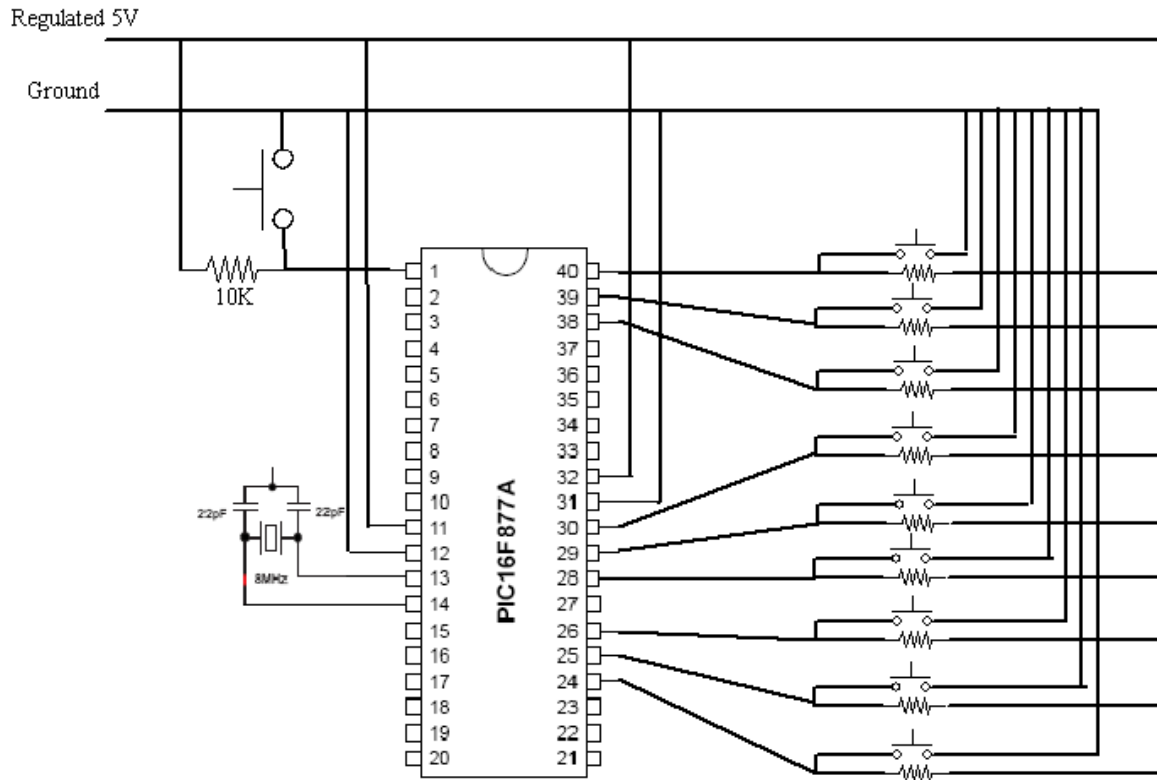
**Figure 4.10: The Controller PCB**

## **4.5 Sensor system**

This robotic arm has a small sensor system that is used to get the status of the grasping process. It has 9 tiny push-buttons located in the contact area of each finger (3 for each finger). Each push-button is connected directly with a pin in the PIC and the ground. So when pressed, all the current will flow through it to the ground leaving the pin of the PIC with logical voltage “0”. Thus, the number of pressed push-buttons is known.

### **4.5.1 Grasping Controller**

The schematic of the controller is shown in Figure 4.11:



**Figure 4.11: Grasping Controller**

The grasping controller also uses a Microchip PIC 16F877 microcontroller. It has the same configuration as the main controller. However, the port B (pins 38-40), port C (pins 24-26) and port D (pins 28-30) are connected to push-buttons and resistors just like the circuit of the hardware reset connected to pin 1. These 9 push-button are placed in the fingers (3 per finger) and are used to know the status of grasping. Its output is directed to port C (pins 16-18) to be transmitted to the main controller through its port D (pins 28-30).

# Chapter 5

## System Integration – Software

### 5.1 Introduction

The Software part of this thesis is composed of 3 main components: The grasping controller, the main controller and the graphical user interface that will control the movements of the robot and gives feedback about the grasping state and the connectivity with the robotic arm. This chapter will detail the design and the implementation of these 3 components as well as related topics (e.g. RS-232 communication protocol and PWM for servo control).

### 5.2 Grasping Controller

As discussed earlier (section 4.5.1), the grasping circuit consists of 9 push-buttons that acts as a sensor system for the robotic arm to sense the grasping status. These push-buttons will be pressed by the objects when they are grasped. According to the number and the position of the pressed buttons, this controller will detect the status of the grasped object. It has 3 statuses to be outputted to the main controller:

- **UNGRASPED:** This will be the case when the object is totally not touching the fingers or it is partially touching a finger on one side only.
- **PARTIALLY\_GRASPED:** This will be the case when the object is partially touching on the two sides of the fingers.

- **GRASPED:** This will be the case when the object is totally touching the fingers.

From the push-buttons point of view:

Let the group of push-buttons on the left finger called L1, the upper group of push-buttons on the right finger called R1 and the lower group of push-buttons on the right finger called R2. The '+' is the algebraic sum of two numbers, the '!=' means “not equal” and the '>=' means “greater than or equal”. The numbers in the equation represents the number of push-buttons that are pressed. For example, L1=3 means that there is 3 pressed push-buttons on the left finger). The output will be according to the following rules:

- **UNGRASPED:** ( $L1 = 0$  or ( $R1 = 0$  and  $R2 = 0$ )).
- **PARTIALLY\_GRASPED:** ( $L1 \neq 0$  and ( $R1+R2 \neq 0$ )).
- **GRASPED:** ( $L1 \geq 2$  and ( $R1+R2 \geq 3$ )).

Since there are 9 push-buttons, each one is either opened or closed and they are independent on each other, therefore there are 512 cases. The above rules will result distribute these cases according to Table 5.1:

**Table 5.1: Grasping status**

|                          | <b>No of cases</b> | <b>percentage</b> |
|--------------------------|--------------------|-------------------|
| <b>UNGRASPED</b>         | 71                 | 13.87             |
| <b>PARTIALLY_GRASPED</b> | 273                | 53.32             |
| <b>GRASPED</b>           | 168                | 32.81             |

The controller will have to check out the status starting with the “grasped” state then the “partially\_grasped” state and finally the “ungrasped” state to avoid the overlapping statuses.

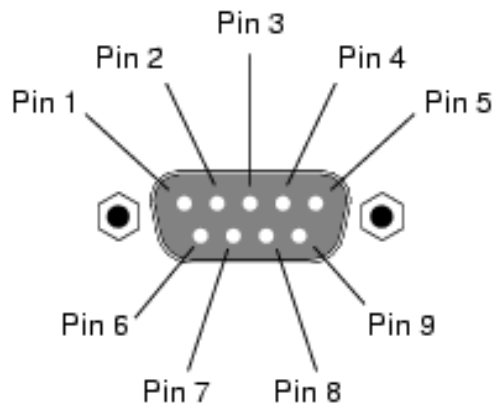
### 5.3 Main Controller

The aim of the main controller is to make the communication with the graphical user software through the serial port and to position the servos accordingly. First, it has to connect to the PC

Software. Then, it asks the PC Software for the value of each servo and updates that servo accordingly.

### 5.3.1 The RS-232 Communication protocol

Recommended Standard 232 is a standard used for serial communication between 2 entities. Usually, these 2 entities are called the Data Terminal Equipment (DTE) which is the computer and the Data Circuit-terminating Equipment (DCE) which is the device. To send some data between the 2 entities, the data must be represented by a set of bytes. These bytes are sent on the communication link one after another. The serial communication means that the 1 byte is sent as 8 bits one after another. This implies that only one wire is needed in the communication. On opposite side, the parallel communication means that the 8 bits of the 1 byte are sent simultaneously over 8 wires. Clearly, the parallel communication is much faster than the serial one. For this thesis, the speed of the serial communication was more than enough for the data transfer. The DB9 connector was selected due to its popularity (Figure 5.1).



**Figure 5.1: DB9 male connector**

The Pinouts are distributed as follows:

1. **Data Carrier Detect (DCD):** The DCE sends to the DTE when the connection is established successfully.
2. **Received Data (RxD):** The actual data line from the DCE to the DTE.

3. **Transmitted Data (TxD):** The actual data line from the DTE to the DCE.
4. **Data Terminal Ready (DTR):** Used by the DTE to inform the DCE that it is ready for communication.
5. **System Ground (GND):** the common ground for the DTE and the DCE.
6. **Data Set Ready (DSR):** Used by the DCE to inform the DTE that it is ready for communication (it is powered on).
7. **Request To Send (RTS):** Used by the DTE to prepare the DCE to receive data. It is mainly used in hardware flow control.
8. **Clear To Send (CTS):** Used by the DCE to acknowledge the RTS signal sent by the DTE. It is mainly used in hardware flow control.
9. **Ring Indicator (RI):** Used in the modems devices to inform the DTE when it detects a ring signal from the telephone line. It is not connected here.

Commonly, there are 2 types of connections for the serial port:

#### 1. 3-wire

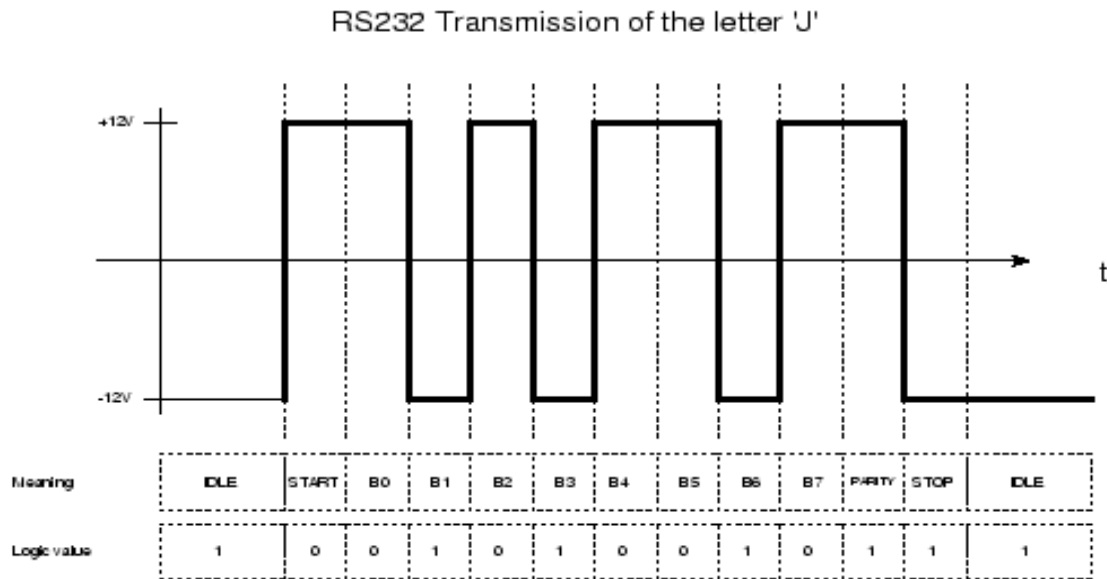
In this mode, only 3 wires are connected. Pins 2 and 3 for the send/receive channels. Pin 5 for the ground. If the communication is done in one direction only, a data channel (Pin 2 or 3) may be disconnected. This is the mode used in this project.

#### 2. 5-wire

In addition to the connections of the 3-wire mode, the Pins 7 and 8 for the hardware flow control may be connected.

As discussed earlier, the byte is sent by sending the bits one after another. This is shown in Figure 5.2.

Note that the logical values used in this protocol are inverted to the voltage values. This means that, the logical zero is +12V and logical one is -12V.



**Figure 5.2: Transmission of a byte in RS232 (from: [www.best-microcontroller-projects.com](http://www.best-microcontroller-projects.com))**

The transmission is done by getting the binary form of the character to be sent from the ASCII table. The ASCII “American Standard Code for Information Interchange” table is a set of commonly used characters (about 128 characters). They are distributed as follows:

- 0-31, 127: Control Codes
- 32-47, 58-64, 81-86, 123-126: Punctuation and mathematical symbols
- 48-57: Numbers 0-9
- 65-80: Alphabetical Capital letters A-Z
- 87-122: Alphabetical Small letters a-z

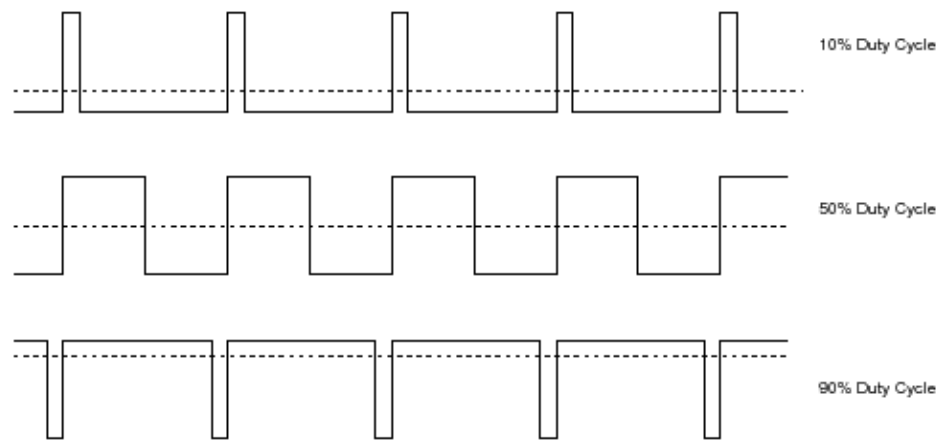
After that conversion, the transmitter prepares a packet (as shown in Figure 5.2) and assigns the data bits to the bits B0 to B7. A parity bit may be used to check the validity of the received data. It may be even/odd parity. This is a very simple method that can count the number of logical ones in the data byte and validate according to the parity bit. It is not used in this project, as it is not expected to have much noise in the environment surrounding the robot. Finally, the entire packet is enclosed between a “START” and “STOP” bits used for the asynchronous

communication. When the receiver gets the “START” bit, it will start decoding the packet. Simply, the “STOP” signal may be 1, 1.5 or 2 bits and means that the packet is done.

Another important parameter in the RS-232 protocol is the baud-rate. It is the speed of the transmission of the bytes. It is measured in bits per second. From another view, it defines the period of time used to signal each bit. In this project, the chosen baud rate is 2400 bit per second. This means that each bit will consume  $1/2400 = 416\mu s$  to be transferred.

### 5.3.2 Pulse Width Modulation

It is the term used to describe the process of outputting an analogue voltage (i.e. of any value between 0 and 5V). This is used to control the power that is driven to the motor. The main idea is that it uses a square wave of repeated pulses. The pulse width can be changed to change the average value of the whole wave. Looking at the extreme cases, when the square wave has repeated pulses of width 100% of the period, the output is 5V. On the other hand, having the square wave with repeated pulses of width 0% of the period, the output is 0V. So, to get an average value of 2.5V, we will have to get a square wave of repeated pulses, where each pulse has a width of 50% of the period. More examples are shown in Figure 5.3.



**Figure 5.3: PWM (from: [www.best-microcontroller-projects.com](http://www.best-microcontroller-projects.com))**

The ratio of the width of the pulse to the period length is called duty cycle. As shown in Figure 5.3, the dotted line represents the output average value. This value can vary between 0 and 5V according to the value of the duty cycle.



There are two ways to generate that effect. First way is to use some analogue circuits to generate the wave. That way is expensive and not very effective as the analogue components will drift after a while. The other approach is to use the PIC microprocessor to generate the wave. This way is more stable and cheap as it does not require extra hardware. Also, this generated signal will not drift as it is generated from the time base of the microprocessor (i.e. crystal).

To generate the wave using the second approach, it is required to know the frequency needed. Then, it is easy to get the period length using the relation:

$$\text{Period} = 1 / \text{Frequency}$$

Knowing the period length, it is required to divide that length into an “on” part and an “off” part such that the ratio of the “on” part to the period length is equal to the needed duty cycle. Finally, the code can be written as shown in Table 5.2.

**Table 5.2: PWM code**

```
//let PIN is the pin where the output is obtained.  
//let t1 is the “on” part of the pulse and t2 is the “off” part.  
//Delay_ms(time) is a function to wait for amount of time.  
While(true){  
    PIN = 1  
    Delay_ms(t1)  
    PIN=0  
    Delay_ms(t2)  
}
```

The idea here is very simple. The pulses will be generated continuously. Each pulse will start with the “on” part and then switches to the “off” part. The addition of these two parts will results in the period length which when repeated, the required frequency will be obtained. And as discussed earlier, the duty cycle that is used in the calculations will directly affect the output average value.

### 5.3.2.1 PWM for Servos

Servos use the PWM signals to position their output shaft. The exact information on the PWM signal of a specific motor varies according to the type of the motor. However, most of the servos available now are similar to each other. They use a frequency of 50Hz. That means, the period length is  $1/50 = 20$  msec. Within this period, the motor is concerned only with the “on” part of the signal. The width of the pulses used directly moves the output shaft to a certain position as shown in Figure 5.4.

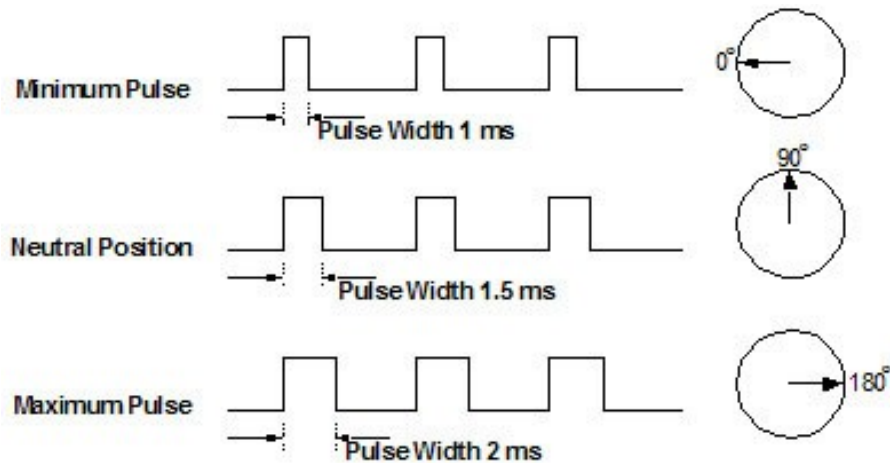


Figure 5.4: PWM for servo motors (from: [www.societyofrobots.com](http://www.societyofrobots.com))

As shown, having a pulse width of 1ms will position the motor on its extreme left endpoint. On the other side, a pulse width of 2ms will position the motor on its other extreme endpoint. Again, the exact timings depend on the servo motor itself. However, the same idea is applicable for all servos.

There are 3 wires to control the servo. Two of them are directly connected to the battery source while the third is used by the microcontroller to output its PWM signal to the servo motor. Refer to Figure 5.5 for the colors of widely used servo motors.

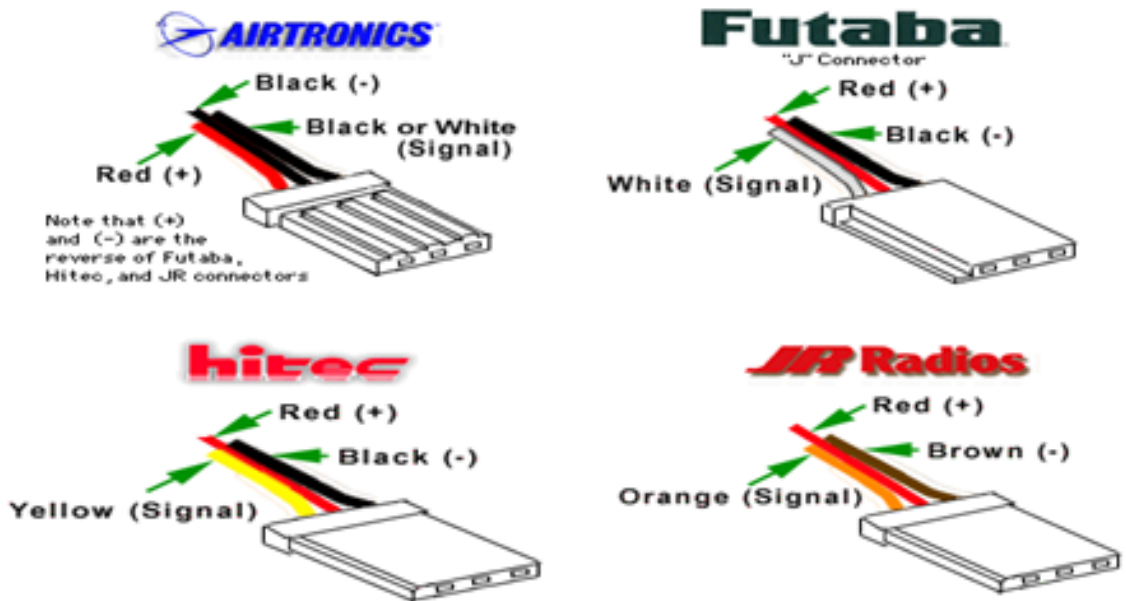


Figure 5.5: Wiring Colors for widely used servos (from: [www.fatlion.com](http://www.fatlion.com))

### 5.3.3 Main Controller code

The C code used for the main controller depends heavily on the 2 previous topics, RS-232 communication and the PWM signals. A general idea for each function in the code is shown in Table 5.3.

Table 5.3: Main Controller Code

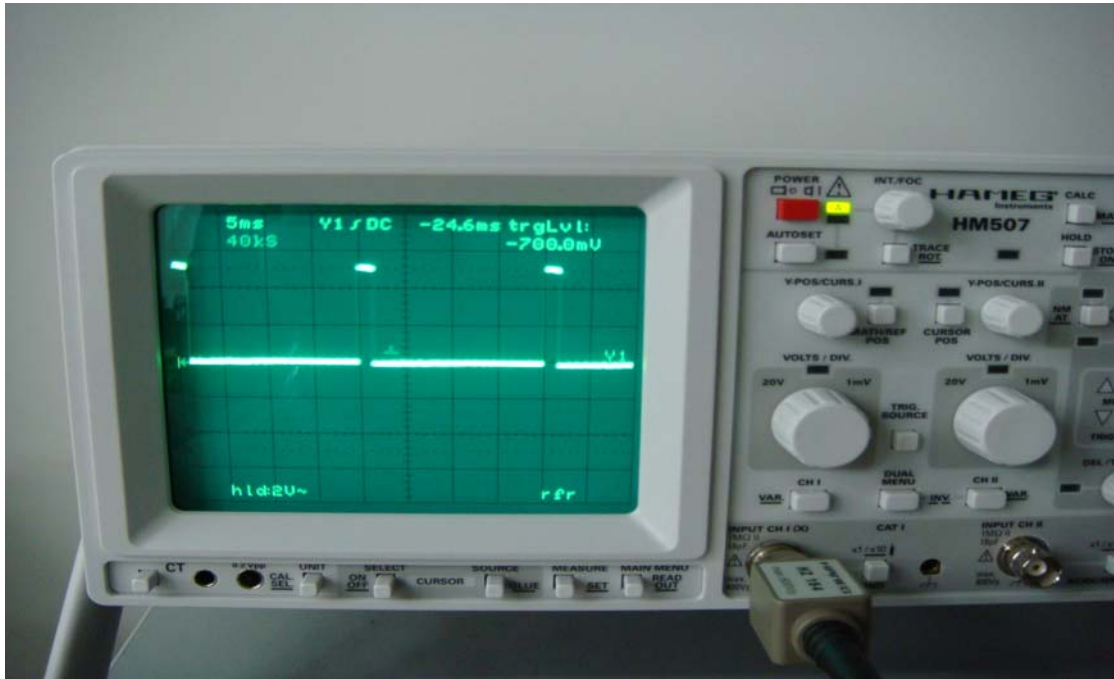
```
void PC_Connect(){
//this function will use the RS-232 serial port to send and receive a simple
handshaking messages. It is a blocking function; this means that the code
will not continue after this the calling of this function except when the
connection has been created successfully. The protocol specification is very
simple. The microcontroller will send "Welcome" message to the PC and
then waits for "Hello" message from the PC. However, if the PC sends
incorrect message, the microcontroller will re-start the protocol and sends
"Welcome" message to the PC.
}

int Read_Data(){
//this function will use the RS-232 serial port to receive the pulse width
value of the current servo to be updated from the PC. Converting that value
```

```

to integer data type to easily manipulate it later.
}
void My_Delay(int t){
//this is a small helper function that is used to wait for a “t” us. The behavior
of this function was monitored by the oscilloscope to accurately get its
execution time (see Figure 5.6).
}
void Generate_Pulse(){
//this function will use the PWM signals to position the current servo motor
according to the value read by the Read_Data() function.
}
void Fingers(){
//this function will use the RS-232 serial port to output the status got from
the grasping controller to the PC.
}
void main(){
//this is the main function. It first initializes the declared variables and
declares that PORTB is an output port (servo motor control signals). It then
enters an infinite loop to update the position of all the servos in sequence.
For each servo, the function Read_Data() is used to get the value from the
PC and then the function Generate_Pulse() is used to output that value on
the specified pin in the port. After each update, call the Fingers() function to
output the status received from the grasping controller.
}

```



**Figure 5.6: The oscilloscope test for the output of the `Generate_Pulse()` function.**

**The pulse width was 1500us and the frequency is 50Hz.**

As discussed earlier in the PWM (section 5.3.2.1), the required frequency is 50Hz. This gives a period of 20ms. Knowing the fact that the range of pulse widths for the available servo motors is between 0 and 2ms. and the fact that the servo motors are concerned only with the “on” part of the signal, we are allowed to perform a simple trick to operate all of them. The idea is to make the code concern only with the “on” part of the wave for all the servos. That means, outputting the “on” part of servo#1 and then instead of outputting the “off” part of servo#1, just output a “0” for servo#1 and move to the “on” part of servo#2. In worst case (i.e. all servos need to the extreme endpoint of 2ms), then we have  $8*2=16\text{ms}$  period. That is less than our 20ms period. The overhead of the communication and moving from 1 servo to another may consume that 4ms difference.

## 5.4 PC Software

Figure 5.7 shows the graphical user interface that the user will use to move the robotic arm. It is written in Java. To communicate with the serial port, the Java Communications API (known as

the javax.comm.\* package). This package facilitates the communication between java applications and the available ports on the system.

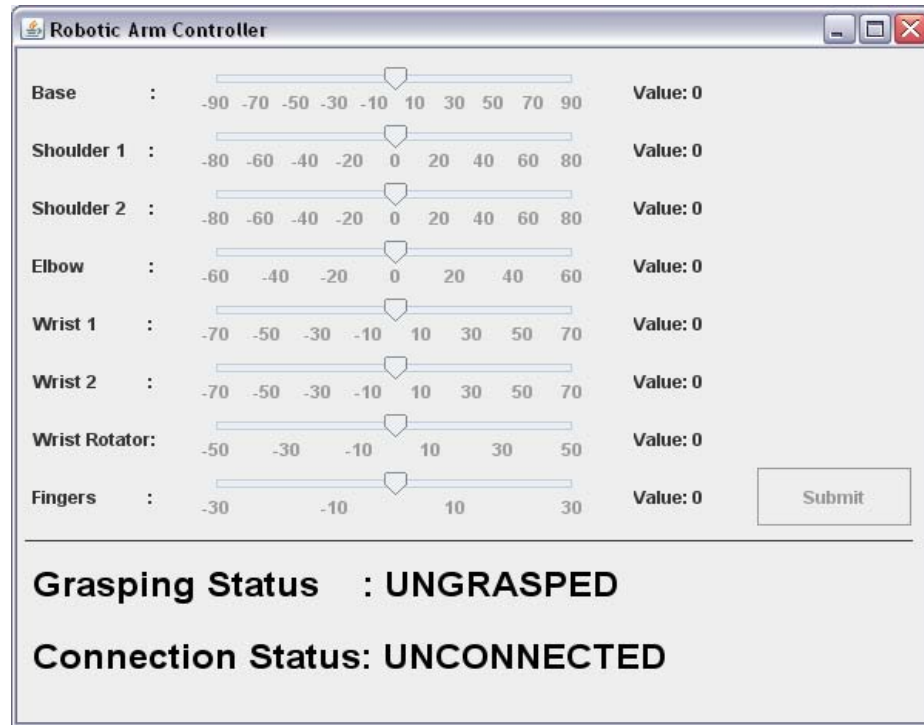


Figure 5.7: PC Software (unconnected)

The GUI is very simple as it contains a table of 8 rows and 4 columns. The 8 rows represent the 8 available servos in the robotic arm (each servo motor is labeled after the joint it represents). Column 1 is JSlider component from the javax.swing package. It represents the extreme end-points that the corresponding servo can rotate to. It has a small slider that can be used to change the current angle for each servo. The third column represents the current angle for each servo. Finally, the fourth column consists of a single JButton that is used to update all the servos. This will allow the user to change many servos and send the final status of all servos once to the microcontroller.

Under that table, there are 2 labels that identify the current grasping state as reported by the main controller which got it from the grasping controller. The other is needed to know whether the robotic arm controller is connected to the PC or not.

While the micro-controller is unconnected to the PC, the 8 sliders will be de-activated to prevent the user from changing their values. The same applies for the “submit” button, it is de-activated. Once the microcontroller is connected, they will be enabled as shown in Figure 5.8.

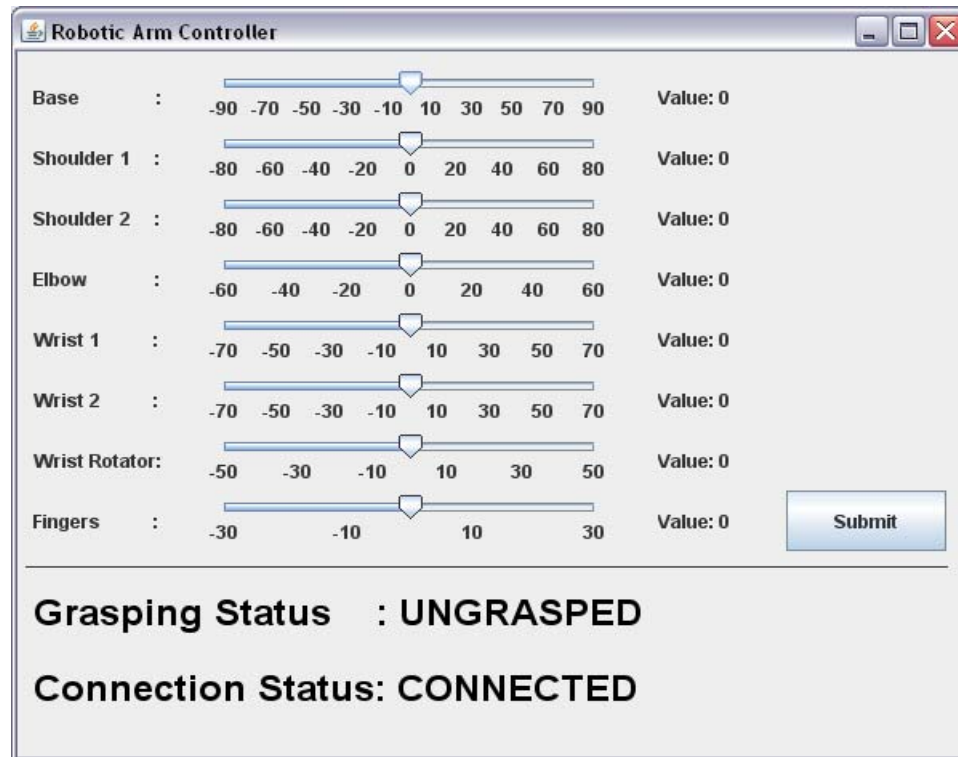


Figure 5.8: PC Software (connected)

# Chapter 6

## Conclusion and Future Work

### 6.1 Conclusion

Throughout the chapters of this thesis, many topics were introduced and discussed such as:

- Geometrical Classification of Robots
- Applications of Industrial Robots
- Morphology of Robotic Arms
- Usage of Microchip PIC 16F877 microcontroller
- RS-232 Communication Protocol
- Servo Motors
- Pulse Width Modulation
- Using javax.comm package (to connect java application with the serial port)

These topics were used in the design and the implementation of a 6-DOF Robotic Arm that can be used in a variety of environments and for different purposes. The main consideration was on the Industrial Robotics. Different geometrical designs for Robotic Arms were presented and the Jointed Design was selected to be implemented. Different types of Production tooling were presented and the grasping robotic hand was selected to be implemented. A hardware controller



was designed to control the 8 used servos and at the same communicate with the PC software through RS-232 communication protocol. A separate mini-controller was designed to sense the status of the grasping at the hand. Finally, a PC software that has a graphical user interface was designed and implemented that is capable of setting the position of the 8 servos used in the robot.

## **6.2 Future Work**

The robotic arm was designed to load heavy weights and to be accurate. But that needs servos that have high torque and are accurate. Unfortunately, these servos were not available at the local market at the time of working in this thesis. So it is strongly recommended to consider this design (mechanical part only) as a prototype to describe the functionality and use servos of high torque (not less than 25 Kg.cm) in real industrial robots.

Another main functionality that needs to be considered is the use of Kinematics calculations. This is an update to the PC software to ease the life of the user. It will allow the user the opportunity to visualize the environment surrounding the robot as 3D space. The user will be able to select a point in that space without worrying about the servos configurations that will deliver the production tool to that point. That's the role of the PC software to take as an input the 3 co-ordinates of the desired point and then automatically adjust the 8 servos to reach that point.

A third functionality to be added is that the ease of manually positioning the robot using teach pendant for example. To do that, the robot must be able to easily know its current configuration of servos using feedback sensors. The user can position the arm in the desired target position and then read that configuration from the PC software or the teach pendant. This functionality permits programming the robot by teaching or guidance to do some task.

## Appendix A Servo Motors

This thesis uses 3 types of servos as shown in Table A.1:

**Table A.1: Used Servos**

| Futaba S3003 Servo<br>Standard                                                      | Hitec HS-5645MG High-<br>Torque 2BB                                                 | GWS S125 1T BB Sail<br>Winch                                                          |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |

Table A.2 is a comparative study between different types of servos. It presents 2 analog servos of low torque. Also 3 analog and 1 digital servos of high torque are presented in Table A.3.

**Table A.2: Low Torque Servos**

|                           | <b>Hitec HS-311 Servo Standard</b> | <b>Futaba S3003 Servo Standard</b>                                                                                                                        |
|---------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analog vs. Digital</b> | Analog                             | Analog                                                                                                                                                    |
| <b>Ball Bearing</b>       | non-ball bearing                   | non-ball bearing (can be added manually by an additional ball-bearing kit, or using “Futaba S3004 Servo” which is the ball bearing version of this servo) |
| <b>Gear Type</b>          | Nylon                              | Nylon                                                                                                                                                     |
| <b>Motor Type</b>         | 3-pole                             | 3-pole                                                                                                                                                    |
| <b>Speed @ 4.8V</b>       | 0.19 sec/60°                       | 0.23 sec/60°                                                                                                                                              |
| <b>Speed @ 6.0V</b>       | 0.15 sec/60°                       | 0.16 sec/60°                                                                                                                                              |
| <b>Torque @ 4.8V</b>      | 42 oz-in                           | 44 oz-in                                                                                                                                                  |
| <b>Torque @ 6.0V</b>      | 51 oz-in                           | 56.8 oz-in                                                                                                                                                |
| <b>Length</b>             | 41 mm                              | 41 mm                                                                                                                                                     |
| <b>Width</b>              | 20 mm                              | 20 mm                                                                                                                                                     |
| <b>Height</b>             | 37 mm                              | 36 mm                                                                                                                                                     |
| <b>Weight</b>             | 43 gram                            | 37.2 gram                                                                                                                                                 |
| <b>Price</b>              | 8.99 \$                            | 10.99 \$                                                                                                                                                  |
| <b>My Usage</b>           | No                                 | Used 4 in the hand and the wrist                                                                                                                          |

**Table A.3: High Torque Servos**

|                                   | <b>Hobbico CS-80<br/>2BB Giant Scale<br/>Servo</b> | <b>Hitec HS-5645MG<br/>High-Torque 2BB</b> | <b>GWS S125 1T BB<br/>Sail Winch</b> | <b>HS-815BB<br/>Mega Sail Arm</b> |
|-----------------------------------|----------------------------------------------------|--------------------------------------------|--------------------------------------|-----------------------------------|
| <b>Analog<br/>vs.<br/>Digital</b> | <b>Analog</b>                                      | <b>Digital</b>                             | <b>Analog</b>                        | <b>Analog</b>                     |
| <b>Ball<br/>Bearing</b>           | Dual Ball<br>Bearings                              | Dual Ball Bearings                         | Dual Ball Bearings                   | Dual Ball<br>Bearings             |
| <b>Gear<br/>Type</b>              | Enormous Gear<br>Train                             | Three metal gears                          | Nylon                                | All Heavy Duty<br>Nylon Gears     |
| <b>Motor<br/>Type</b>             | Cored Motor<br>Design                              | 3 Pole Ferrite                             |                                      | 3 Pole Ferrite                    |
| <b>Speed @<br/>4.8V</b>           | 0.19 sec                                           | 0.23 sec/60°                               | 0.26 sec/60°                         | 0.48 sec/140°                     |
| <b>Speed @<br/>6.0V</b>           | 0.14 sec                                           | 0.18 sec/60°                               | 0.21 sec/60°                         | 0.38 sec/140°                     |
| <b>Torque<br/>@ 4.8V</b>          | 275 oz-in                                          | 143.03 oz-in                               | 92 oz-in                             | 274.96 oz-in                      |
| <b>Torque<br/>@ 6.0V</b>          | 343 oz-in                                          | 164.2 oz-in                                | 106 oz-in                            | 343.01 oz-in                      |
| <b>Length</b>                     | 66 mm                                              | 40.6 mm                                    | 40.5 mm                              | 66 mm                             |
| <b>Width</b>                      | 33 mm                                              | 19.8 mm                                    | 20 mm                                | 30 mm                             |
| <b>Height</b>                     | 57.6 mm                                            | 37.8 mm                                    | 42 mm                                | 57.6 mm                           |
| <b>Weight</b>                     | 152 gram                                           | 60 gram                                    | 50 gram                              | 152 gram                          |
| <b>Price</b>                      | 29.99 \$                                           | 49.99 \$                                   | 18.75 \$                             | 48.99 \$                          |
| <b>My<br/>Usage</b>               | No                                                 | 3 in the shoulder and<br>elbow             | 1 in the base                        | No                                |

## Appendix B Budget Summary

The Body structure of the Robotic Arm costs:

**Table B.1: Body Structure Costs**

| <b>Element</b>                         | <b>Price in LE</b>                 | <b>Quantity</b> | <b>Sub-total</b> |
|----------------------------------------|------------------------------------|-----------------|------------------|
| <b>Aluminum Sheet<br/>100x50x0.1cm</b> | 34                                 | 1               | 34               |
| <b>Wooden box</b>                      | 35                                 | 1               | 35               |
| <b>Bearings</b>                        | 27                                 | 1               | 27               |
| <b>Screws and locking<br/>nuts</b>     | 0.2                                | 80              | 16               |
| <b>Aluminum sheet<br/>machinery</b>    | Done by GUC in the industrial park |                 |                  |
| <b>Aluminum sheet<br/>Assembly</b>     | Done by me in the workshop         |                 |                  |

Total Cost for the body structure = 112 LE

Servo Motors costs:

**Table B.2: Servo Motors Costs**

| <b>Element</b>                 | <b>Price in LE</b> | <b>Quantity</b> | <b>Sub-total</b> |
|--------------------------------|--------------------|-----------------|------------------|
| <b>Hitec HS-5645MG<br/>2BB</b> | 415                | 3               | 1245             |
| <b>Futaba S3003</b>            | 89                 | 4               | 356              |
| <b>GWS S125 Winch</b>          | 132                | 1               | 132              |

Total Cost for Servo Motors = 1733 LE

Controller circuit costs:

**Table B.3: Controller Circuit Costs**

| <b>Element</b>                       | <b>Price in LE</b> | <b>Quantity</b> | <b>Sub-total</b> |
|--------------------------------------|--------------------|-----------------|------------------|
| <b>Microcontroller<br/>p16f877a</b>  | 40                 | 2               | 80               |
| <b>Microcontroller<br/>socket 40</b> | 2                  | 2               | 2                |
| <b>IC-MAX232</b>                     | 12                 | 1               | 12               |
| <b>Capacitor 10uf</b>                | 0.35               | 4               | 1.4              |
| <b>Microcontroller<br/>socket 16</b> | 0.35               | 1               | 0.35             |
| <b>Capacitor 100nf</b>               | 0.25               | 1               | 0.25             |
| <b>IC 7805</b>                       | 1.5                | 1               | 1.5              |
| <b>Diode IN4004</b>                  | 0.15               | 1               | 0.15             |

|                                                    |      |    |      |
|----------------------------------------------------|------|----|------|
| <b>Capacitor 22pf</b>                              | 0.35 | 2  | 0.7  |
| <b>Resistor 10k ohm</b>                            | 0.1  | 11 | 1.1  |
| <b>Crystal 8MHz</b>                                | 4    | 1  | 4    |
| <b>DB9 serial port</b>                             | 3.75 | 1  | 3.75 |
| <b>Green pluggable jack</b>                        | 1    | 1  | 1    |
| <b>DC adapter socket</b>                           | 1.75 | 1  | 1.75 |
| <b>Push-button</b>                                 | 0.75 | 10 | 7.5  |
| <b>Pin header</b>                                  | 3    | 1  | 3    |
| <b>Printed board</b>                               | 2    | 1  | 2    |
| <b>Acids</b>                                       | 1    | 1  | 1    |
| <b>9-volt battery socket</b>                       | 1.25 | 1  | 1.25 |
| <b>Drawing schematic and resolving the bridges</b> | 50   | 1  | 50   |
| <b>Printing the PCB</b>                            | 30   | 1  | 30   |
| <b>Soldering the components</b>                    | 20   | 1  | 20   |

Total cost for the controller: 224.7 LE

**Total Cost of the Robot: 2069.7 LE**

## Appendix C TRUMPH Machine Report

This is the report generated by the “TRUMPH” machine while the cutting of the Aluminum sheet to construct the body of the robotic arm.

Note that, the report refers to the pieces with different names!

**TRUMPF**



### SET-UP SCHEDULE GENERAL DATA

<<Private\_Info>>  
07.04.2008  
ToPs 300 V5.10.0

|                             |                                        |
|-----------------------------|----------------------------------------|
| MACHINE:                    | TC3000R - TYPE: 1 (2500.0 x 1250.0 mm) |
| CONTROL:                    | Bo PNC-P VARIANT: 1                    |
| COMPANY:                    |                                        |
| JOB NAME:                   |                                        |
| NC-PROGRAM NAME:            | /<<Private_Info>>/RoboticArm.LST       |
| PROGRAM NAME:               | <b>ROBOTICARM</b> (RoboticArm)         |
|                             |                                        |
| MATERIAL ID:                | <b>AlMg3-10 (AlMg3)</b>                |
| STORAGE ITEM DESIGNATION:   |                                        |
| BLANK:                      | <b>500 x 1000 x 1 mm</b>               |
| WEIGHT:                     | 1.35 kg                                |
| MACHINING TIME:             | <b>0 : 03 : 33 [h:min:s]</b>           |
| STORAGE REQUIREMENT:        | 14127 CHARACTERS                       |
| LASER TOTAL CUTTING LENGTH: | 0.000 mm                               |
| NUMBER OF PROGRAMME RUNS:   | 1                                      |
| SCRAP:                      | 59.37 %                                |
|                             |                                        |



| <b>MACHINING INSTRUCTIONS</b> |                                |    |    |
|-------------------------------|--------------------------------|----|----|
| CLAMP POSITIONS:              | 2                              | 12 | 15 |
| LOCATOR PIN:                  | 1 FROM THE LEFT                |    |    |
| STARTING POINT:               | X = 1905.901 mm,Y = 140.010 mm |    |    |
| REMARKS:                      |                                |    |    |

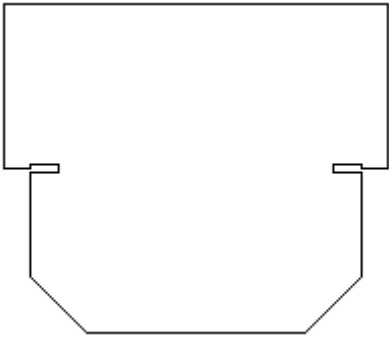
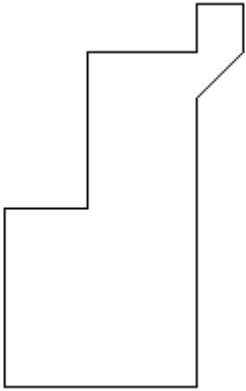
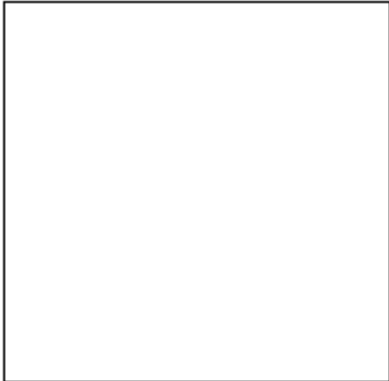
| <b>TOOL LIST</b>                             |      |              |              |              |       |                   |    |    |    |         |
|----------------------------------------------|------|--------------|--------------|--------------|-------|-------------------|----|----|----|---------|
| IDENT NO                                     | TYPE | DIMENSIO N 1 | DIMENSIO N 2 | DIMENSIO N 3 | ANGLE | REMARK            | WT | MT | LT | STROKES |
| 101999006                                    | 3    | 2.000        | 0.000        | 0.000        | 0.000 | 1/10:Squar 2.0mm  | 0  | 10 | 0  | 80      |
| 04150030                                     | 4    | 15.000       | 3.000        | 0.000        | 0.000 | Recta 15.0x 3.0mm | 1  | 1  | 0  | 44      |
| 04200050                                     | 4    | 20.000       | 5.000        | 0.000        | 0.000 | Recta 20.0x 5.0mm | 1  | 1  | 0  | 420     |
| <b>NUMBER OF STEEL/RTC CARTRIDGES: 3 / 0</b> |      |              |              |              |       |                   |    |    |    |         |

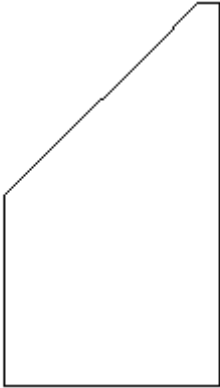
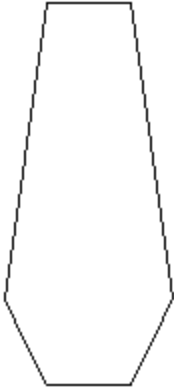
| PTT   | IDENT NO  | REMARK            | TDC   | BDC   | SOFTPUNCH |
|-------|-----------|-------------------|-------|-------|-----------|
| PTT-1 | 101999006 | 1/10:Squar 2.0mm  | 0.000 | 0.000 | 0         |
| PTT-2 | 04150030  | Recta 15.0x 3.0mm | 0.000 | 0.000 | 0         |
| PTT-3 | 04150030  | Recta 15.0x 3.0mm | 0.000 | 0.000 | 0         |
| PTT-4 | 04200050  | Recta 20.0x 5.0mm | 0.000 | 0.000 | 0         |

| <b>INFORMATION ON SINGLE PART</b> |                 |                         |         |
|-----------------------------------|-----------------|-------------------------|---------|
| PART NUMBER:                      | DRAWING NUMBER: | GEOFILE NAME:           | NUMBER: |
| 1                                 | NOID_1          | /<<Private_Info>>/S.GEO | 2       |
| 2                                 | NOID_2          | /<<Private_Info>>/V.GEO | 1       |
| 3                                 | NOID_3          | /<<Private_Info>>/A.GEO | 1       |
| 4                                 | NOID_4          | /<<Private_Info>>/N.GEO | 1       |
| 5                                 | NOID_5          | /<<Private_Info>>/F.GEO | 2       |
| 6                                 | NOID_6          | /<<Private_Info>>/H.GEO | 2       |
| 7                                 | NOID_7          | /<<Private_Info>>/M.GEO | 1       |
| 8                                 | NOID_8          | /<<Private_Info>>/T.GEO | 1       |
| 9                                 | NOID_9          | /<<Private_Info>>/B.GEO | 2       |
| 10                                | NOID_10         | /<<Private_Info>>/K.GEO | 2       |
| 11                                | NOID_11         | /<<Private_Info>>/O.GEO | 2       |

|             |         |                         |   |
|-------------|---------|-------------------------|---|
| 12          | NOID_12 | /<<Private_Info>>/U.GEO | 3 |
| 13          | NOID_13 | /<<Private_Info>>/Q.GEO | 2 |
| 14          | NOID_14 | /<<Private_Info>>/D.GEO | 2 |
| SHEET NAME: |         |                         |   |

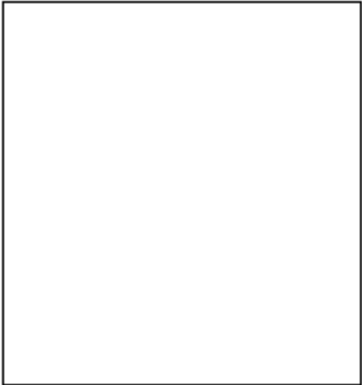
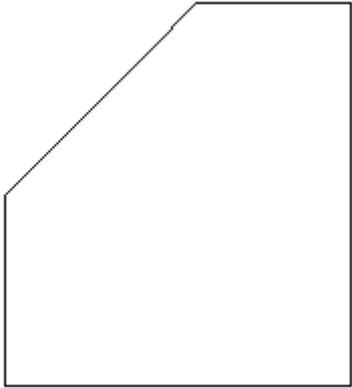
## INFORMATION ON SINGLE PART

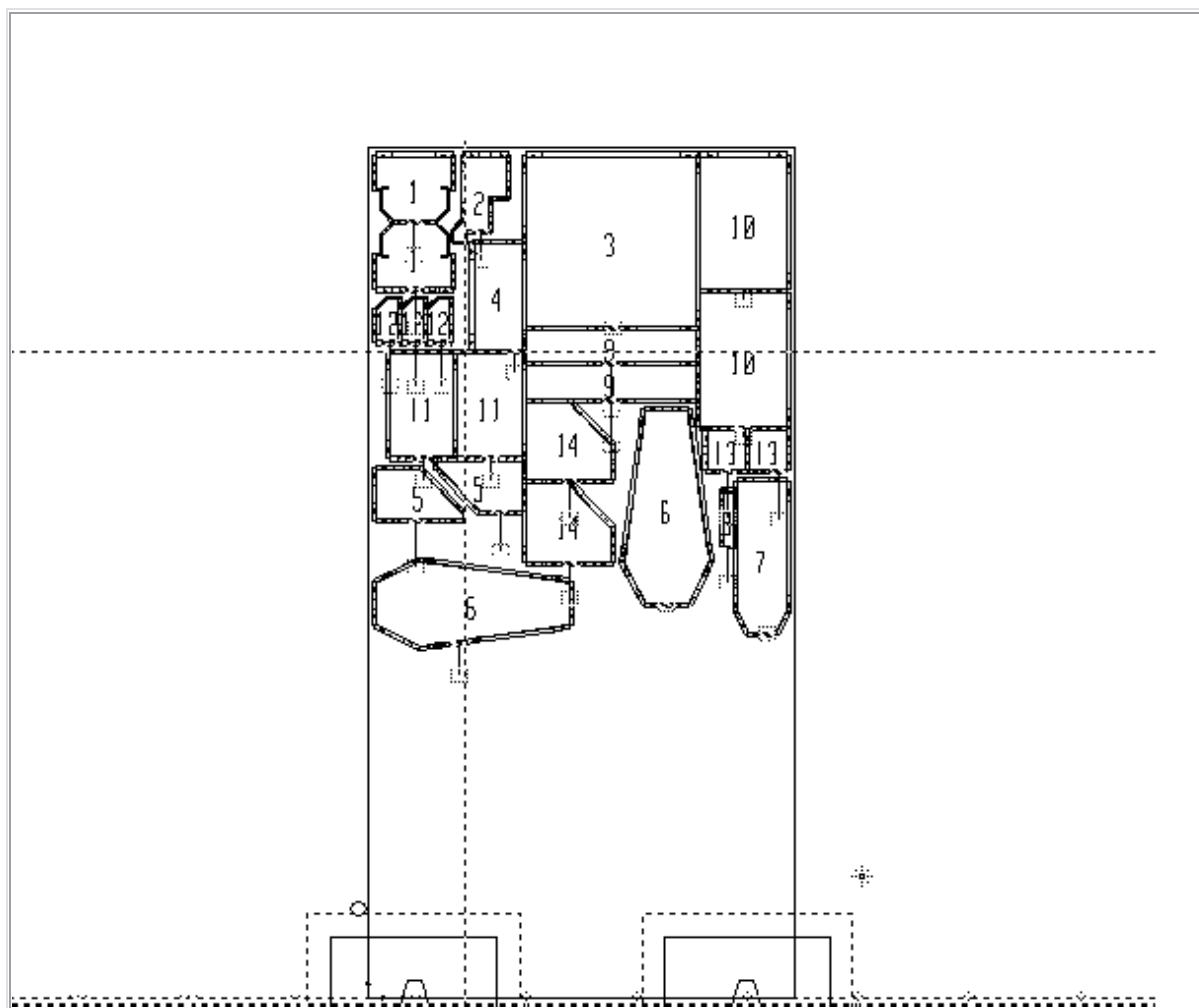
|                                                                                     |                       |                          |
|-------------------------------------------------------------------------------------|-----------------------|--------------------------|
|    | PART NUMBER:          | 1                        |
|                                                                                     | DRAWING NUMBER:       | NOID_1                   |
|                                                                                     | NAME OF DRAWING:      |                          |
|                                                                                     | CUSTOMER NAME:        |                          |
|                                                                                     | NUMBER:               | 2                        |
|                                                                                     | DIMENSIONS:           | 87.500 x 75.000 mm       |
|                                                                                     | SURFACE:              | 5912.50 mm <sup>2</sup>  |
|                                                                                     | WEIGHT:               | 0.016 kg                 |
|                                                                                     | MACHINING TIME:       | 0.27 min                 |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                 |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/S.GEO  |
|                                                                                     |                       |                          |
|   | PART NUMBER:          | 2                        |
|                                                                                     | DRAWING NUMBER:       | NOID_2                   |
|                                                                                     | NAME OF DRAWING:      |                          |
|                                                                                     | CUSTOMER NAME:        |                          |
|                                                                                     | NUMBER:               | 1                        |
|                                                                                     | DIMENSIONS:           | 62.400 x 99.900 mm       |
|                                                                                     | SURFACE:              | 3718.05 mm <sup>2</sup>  |
|                                                                                     | WEIGHT:               | 0.010 kg                 |
|                                                                                     | MACHINING TIME:       | 0.17 min                 |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                 |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/V.GEO  |
|                                                                                     |                       |                          |
|  | PART NUMBER:          | 3                        |
|                                                                                     | DRAWING NUMBER:       | NOID_3                   |
|                                                                                     | NAME OF DRAWING:      |                          |
|                                                                                     | CUSTOMER NAME:        |                          |
|                                                                                     | NUMBER:               | 1                        |
|                                                                                     | DIMENSIONS:           | 200.000 x 200.000 mm     |
|                                                                                     | SURFACE:              | 40000.00 mm <sup>2</sup> |
|                                                                                     | WEIGHT:               | 0.108 kg                 |
|                                                                                     | MACHINING TIME:       | 0.20 min                 |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                 |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/A.GEO  |
|                                                                                     |                       |                          |
|                                                                                     | PART NUMBER:          | 4                        |

|                                                                                     |                       |                          |
|-------------------------------------------------------------------------------------|-----------------------|--------------------------|
|    | DRAWING NUMBER:       | NOID_4                   |
|                                                                                     | NAME OF DRAWING:      |                          |
|                                                                                     | CUSTOMER NAME:        |                          |
|                                                                                     | NUMBER:               | 1                        |
|                                                                                     | DIMENSIONS:           | 56.250 x 125.000 mm      |
|                                                                                     | SURFACE:              | 7031.25 mm <sup>2</sup>  |
|                                                                                     | WEIGHT:               | 0.019 kg                 |
|                                                                                     | MACHINING TIME:       | 0.13 min                 |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                 |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/N.GEO  |
|   | PART NUMBER:          | 5                        |
|                                                                                     | DRAWING NUMBER:       | NOID_5                   |
|                                                                                     | NAME OF DRAWING:      |                          |
|                                                                                     | CUSTOMER NAME:        |                          |
|                                                                                     | NUMBER:               | 2                        |
|                                                                                     | DIMENSIONS:           | 56.200 x 100.000 mm      |
|                                                                                     | SURFACE:              | 4370.00 mm <sup>2</sup>  |
|                                                                                     | WEIGHT:               | 0.012 kg                 |
|                                                                                     | MACHINING TIME:       | 0.13 min                 |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                 |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/F.GEO  |
|  | PART NUMBER:          | 6                        |
|                                                                                     | DRAWING NUMBER:       | NOID_6                   |
|                                                                                     | NAME OF DRAWING:      |                          |
|                                                                                     | CUSTOMER NAME:        |                          |
|                                                                                     | NUMBER:               | 2                        |
|                                                                                     | DIMENSIONS:           | 100.000 x 225.000 mm     |
|                                                                                     | SURFACE:              | 16875.00 mm <sup>2</sup> |
|                                                                                     | WEIGHT:               | 0.046 kg                 |
|                                                                                     | MACHINING TIME:       | 0.18 min                 |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                 |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/H.GEO  |
|  | PART NUMBER:          | 7                        |
|                                                                                     | DRAWING NUMBER:       | NOID_7                   |
|                                                                                     | NAME OF DRAWING:      |                          |
|                                                                                     | CUSTOMER NAME:        |                          |
|                                                                                     | NUMBER:               | 1                        |
|                                                                                     | DIMENSIONS:           | 56.200 x 175.000 mm      |
|                                                                                     | SURFACE:              | 9522.50 mm <sup>2</sup>  |
|                                                                                     | WEIGHT:               | 0.026 kg                 |

|                                                                                     |                       |                         |
|-------------------------------------------------------------------------------------|-----------------------|-------------------------|
|    | MACHINING TIME:       | 0.15 min                |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/M.GEO |
|                                                                                     |                       |                         |
|    | PART NUMBER:          | 8                       |
|                                                                                     | DRAWING NUMBER:       | NOID_8                  |
|                                                                                     | NAME OF DRAWING:      |                         |
|                                                                                     | CUSTOMER NAME:        |                         |
|                                                                                     | NUMBER:               | 1                       |
|                                                                                     | DIMENSIONS:           | 62.500 x 12.500 mm      |
|                                                                                     | SURFACE:              | 781.25 mm2              |
|                                                                                     | WEIGHT:               | 0.002 kg                |
|                                                                                     | MACHINING TIME:       | 0.13 min                |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/T.GEO |
|                                                                                     |                       |                         |
|  | PART NUMBER:          | 9                       |
|                                                                                     | DRAWING NUMBER:       | NOID_9                  |
|                                                                                     | NAME OF DRAWING:      |                         |
|                                                                                     | CUSTOMER NAME:        |                         |
|                                                                                     | NUMBER:               | 2                       |
|                                                                                     | DIMENSIONS:           | 200.000 x 37.500 mm     |
|                                                                                     | SURFACE:              | 7500.00 mm2             |
|                                                                                     | WEIGHT:               | 0.020 kg                |
|                                                                                     | MACHINING TIME:       | 0.12 min                |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/B.GEO |
|                                                                                     |                       |                         |
|                                                                                     | PART NUMBER:          | 10                      |
|                                                                                     | DRAWING NUMBER:       | NOID_10                 |
|                                                                                     | NAME OF DRAWING:      |                         |
|                                                                                     | CUSTOMER NAME:        |                         |
|                                                                                     | NUMBER:               | 2                       |
|                                                                                     | DIMENSIONS:           | 156.250 x 100.000 mm    |
|                                                                                     | SURFACE:              | 15625.00 mm2            |
|                                                                                     | WEIGHT:               | 0.042 kg                |

|                                                                                     |                       |                         |
|-------------------------------------------------------------------------------------|-----------------------|-------------------------|
|    | MACHINING TIME:       | 0.15 min                |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/K.GEO |
|                                                                                     |                       |                         |
|    | PART NUMBER:          | 11                      |
|                                                                                     | DRAWING NUMBER:       | NOID_11                 |
|                                                                                     | NAME OF DRAWING:      |                         |
|                                                                                     | CUSTOMER NAME:        |                         |
|                                                                                     | NUMBER:               | 2                       |
|                                                                                     | DIMENSIONS:           | 121.800 x 75.000 mm     |
|                                                                                     | SURFACE:              | 9135.00 mm2             |
|                                                                                     | WEIGHT:               | 0.025 kg                |
|                                                                                     | MACHINING TIME:       | 0.13 min                |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/O.GEO |
|                                                                                     |                       |                         |
|  | PART NUMBER:          | 12                      |
|                                                                                     | DRAWING NUMBER:       | NOID_12                 |
|                                                                                     | NAME OF DRAWING:      |                         |
|                                                                                     | CUSTOMER NAME:        |                         |
|                                                                                     | NUMBER:               | 3                       |
|                                                                                     | DIMENSIONS:           | 50.000 x 25.000 mm      |
|                                                                                     | SURFACE:              | 1171.88 mm2             |
|                                                                                     | WEIGHT:               | 0.003 kg                |
|                                                                                     | MACHINING TIME:       | 0.12 min                |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/U.GEO |
|                                                                                     |                       |                         |
|  | PART NUMBER:          | 13                      |
|                                                                                     | DRAWING NUMBER:       | NOID_13                 |
|                                                                                     | NAME OF DRAWING:      |                         |
|                                                                                     | CUSTOMER NAME:        |                         |
|                                                                                     | NUMBER:               | 2                       |
|                                                                                     | DIMENSIONS:           | 43.750 x 46.870 mm      |
|                                                                                     | SURFACE:              | 2050.56 mm2             |
|                                                                                     | WEIGHT:               | 0.006 kg                |
|                                                                                     | MACHINING TIME:       | 0.10 min                |
|                                                                                     | LASER CUTTING LENGTH: | 0.000 mm                |
|                                                                                     | GEOFILE NAME:         | /<<Private_Info>>/Q.GEO |

|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----|-----------------|---------|------------------|--|----------------|--|---------|---|-------------|---------------------|----------|-------------|---------|----------|-----------------|----------|-----------------------|----------|---------------|-------------------------|--|--|--|
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
|  | <table border="1"> <tr> <td>PART NUMBER:</td> <td>14</td> </tr> <tr> <td>DRAWING NUMBER:</td> <td>NOID_14</td> </tr> <tr> <td>NAME OF DRAWING:</td> <td></td> </tr> <tr> <td>CUSTOMER NAME:</td> <td></td> </tr> <tr> <td>NUMBER:</td> <td>2</td> </tr> <tr> <td>DIMENSIONS:</td> <td>90.600 x 100.000 mm</td> </tr> <tr> <td>SURFACE:</td> <td>7810.00 mm2</td> </tr> <tr> <td>WEIGHT:</td> <td>0.021 kg</td> </tr> <tr> <td>MACHINING TIME:</td> <td>0.13 min</td> </tr> <tr> <td>LASER CUTTING LENGTH:</td> <td>0.000 mm</td> </tr> <tr> <td>GEOFILE NAME:</td> <td>/&lt;&lt;Private_Info&gt;&gt;/D.GEO</td> </tr> <tr> <td></td> <td></td> </tr> </table> | PART NUMBER: | 14 | DRAWING NUMBER: | NOID_14 | NAME OF DRAWING: |  | CUSTOMER NAME: |  | NUMBER: | 2 | DIMENSIONS: | 90.600 x 100.000 mm | SURFACE: | 7810.00 mm2 | WEIGHT: | 0.021 kg | MACHINING TIME: | 0.13 min | LASER CUTTING LENGTH: | 0.000 mm | GEOFILE NAME: | /<<Private_Info>>/D.GEO |  |  |  |
| PART NUMBER:                                                                       | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| DRAWING NUMBER:                                                                    | NOID_14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| NAME OF DRAWING:                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| CUSTOMER NAME:                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| NUMBER:                                                                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| DIMENSIONS:                                                                        | 90.600 x 100.000 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| SURFACE:                                                                           | 7810.00 mm2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| WEIGHT:                                                                            | 0.021 kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| MACHINING TIME:                                                                    | 0.13 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| LASER CUTTING LENGTH:                                                              | 0.000 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
| GEOFILE NAME:                                                                      | /<<Private_Info>>/D.GEO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |
|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |    |                 |         |                  |  |                |  |         |   |             |                     |          |             |         |          |                 |          |                       |          |               |                         |  |  |  |



## Appendix D Mechanical System Parts List

The Mechanical structure of the Robotic Arm consists of:

- Body: 19 pieces (see Table D.1).
- Hand: 5 pieces (see Table D.1) & 6 pieces (see Table D.2).
- 2 Bearings:
  1. In the connection between the wooden box and the rotating base.
  2. In the connection between the shoulder and the elbow.

Pieces in Table B1 were cut from an Aluminum sheet of size 100 cm Length, 50 cm Width and 1 mm thickness by the “Trumph” machine. Pieces in Table B2 were supposed to be cut from a sheet of thickness 3 mm to add more grasping power to the fingers. This thickness was not available in the market. So 3 identical pieces for each piece in that table were cut from the 1mm thickness sheet and collected together to form 1 piece of thickness 3 mm.



**Table D.1: Area calculation for 1/16 inch thickness aluminum sheet**

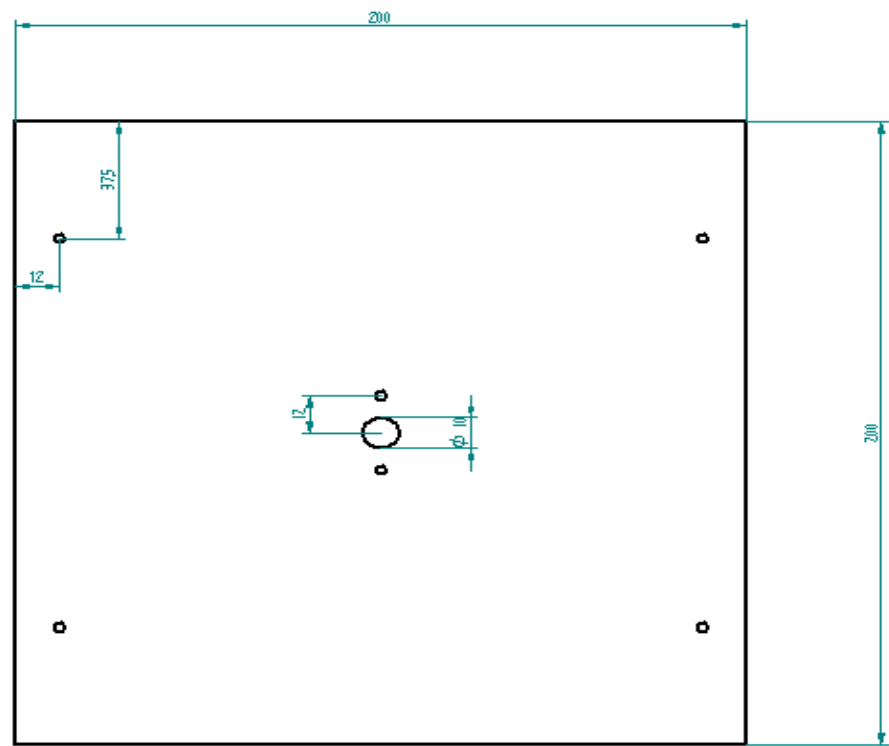
| <b>1/16 inch thick (0.15625 cm)</b> |                    |                   |                                    |                              |                    |
|-------------------------------------|--------------------|-------------------|------------------------------------|------------------------------|--------------------|
| <b>Piece</b>                        | <b>Length (cm)</b> | <b>Width (cm)</b> | <b>Area (cm<sup>2</sup>)/piece</b> | <b>Area (cm<sup>2</sup>)</b> |                    |
| A                                   | 20                 | 20                | 400                                | 400                          |                    |
| B,C                                 | 20                 | 3.75              | 75                                 | 150                          |                    |
| D,E                                 | 9.0625             | 10                | 90.625                             | 181.25                       |                    |
| F,G                                 | 5.625              | 10                | 56.25                              | 112.5                        |                    |
| H,I                                 | 22.5               | 10                | 225                                | 450                          |                    |
| J,K                                 | 15.625             | 10                | 156.25                             | 312.5                        |                    |
| L                                   | 5.625              | 17.5              | 98.4375                            | 98.4375                      |                    |
| M                                   | 5.625              | 12.5              | 70.3125                            | 70.3125                      |                    |
| N,O                                 | 12.1875            | 7.5               | 91.40625                           | 182.8125                     |                    |
| P,Q                                 | 4.6875             | 4.375             | 20.5078125                         | 41.015625                    |                    |
| R,S                                 | 8.75               | 7.5               | 65.625                             | 131.25                       |                    |
| T (Hand)                            | 6.25               | 1.25              | 7.8125                             | 7.8125                       |                    |
| U,V,W (Hand)                        | 5                  | 2.5               | 12.5                               | 37.5                         |                    |
| X (Hand)                            | 10                 | 6.25              | 62.5                               | 62.5                         |                    |
|                                     |                    |                   |                                    |                              |                    |
|                                     |                    |                   | <b>Total</b>                       | 2237.890625                  | (cm <sup>2</sup> ) |
|                                     |                    |                   |                                    | 0.223789063                  | (m <sup>2</sup> )  |

**Table D.2: Area calculation for 1/8 inch thickness aluminum sheet**

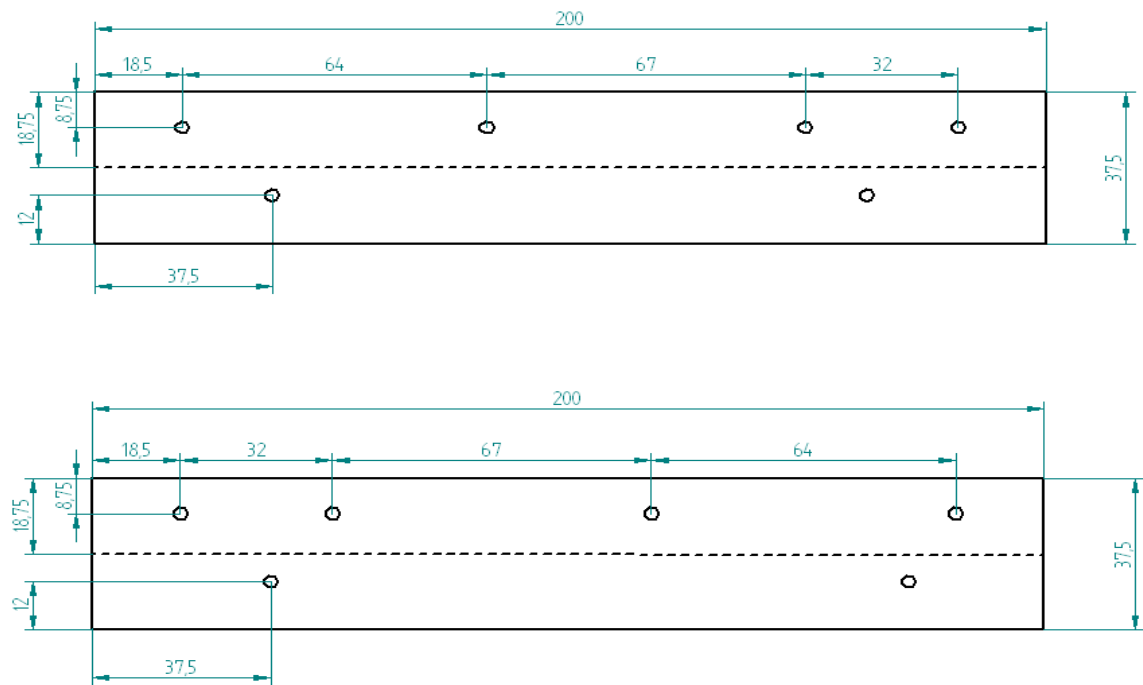
| <b>1/8 inch thick (0.3125 cm)</b> |                    |                                                            |      |
|-----------------------------------|--------------------|------------------------------------------------------------|------|
| <b>1/2 inch width (1.25 cm)</b>   |                    |                                                            |      |
| <b>Piece</b>                      | <b>Length (cm)</b> | <b>Using 3 identical pieces from 0.1 cm thickness (cm)</b> |      |
| A (Hand)                          | 6.25               | 18.75                                                      |      |
| B (Hand)                          | 6.875              | 20.625                                                     |      |
| C (Hand)                          | 9.375              | 28.125                                                     |      |
| D (Hand)                          | 12.5               | 37.5                                                       |      |
| E (Hand)                          | 11.875             | 35.625                                                     |      |
| F (Hand)                          | 6.25               | 18.75                                                      |      |
|                                   |                    |                                                            |      |
| <b>Total</b>                      | 53.125             | 159.375                                                    | (cm) |
|                                   |                    | 1.59375                                                    | (m)  |

The schematic of each piece is shown below:

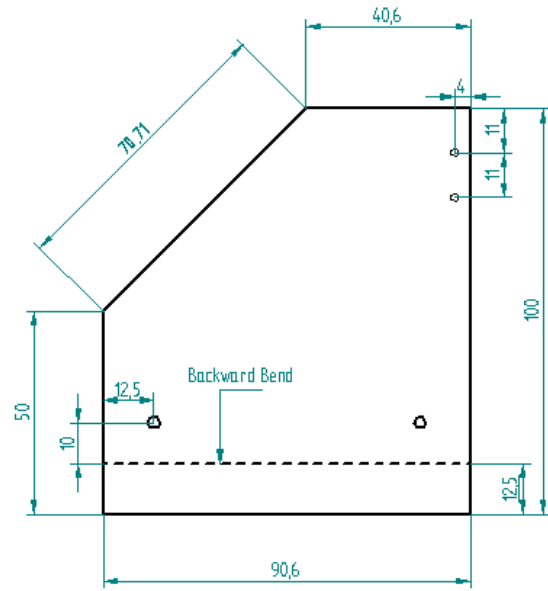
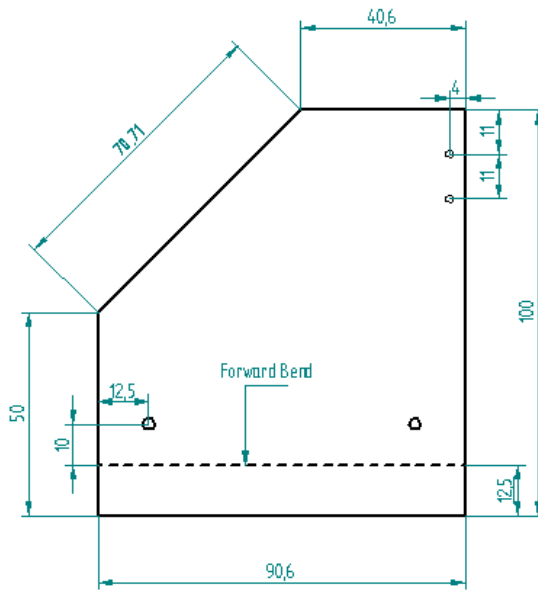
Piece 'A':



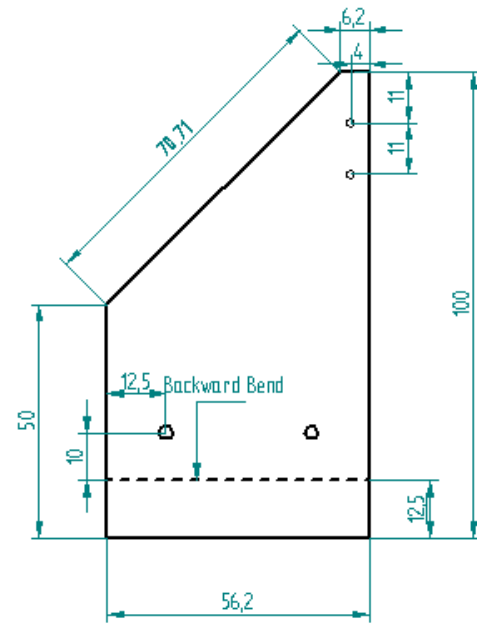
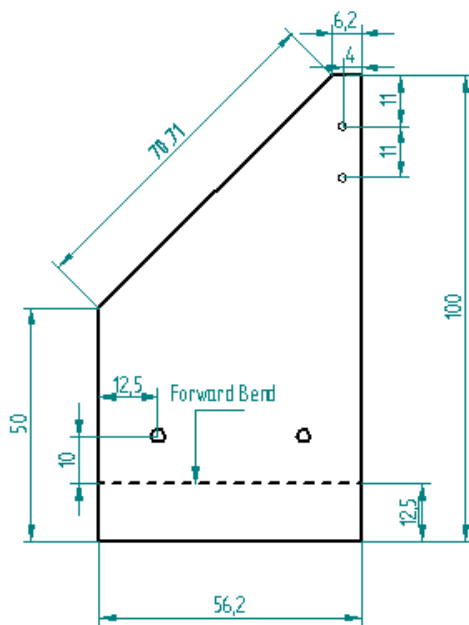
Pieces 'B' & 'C':



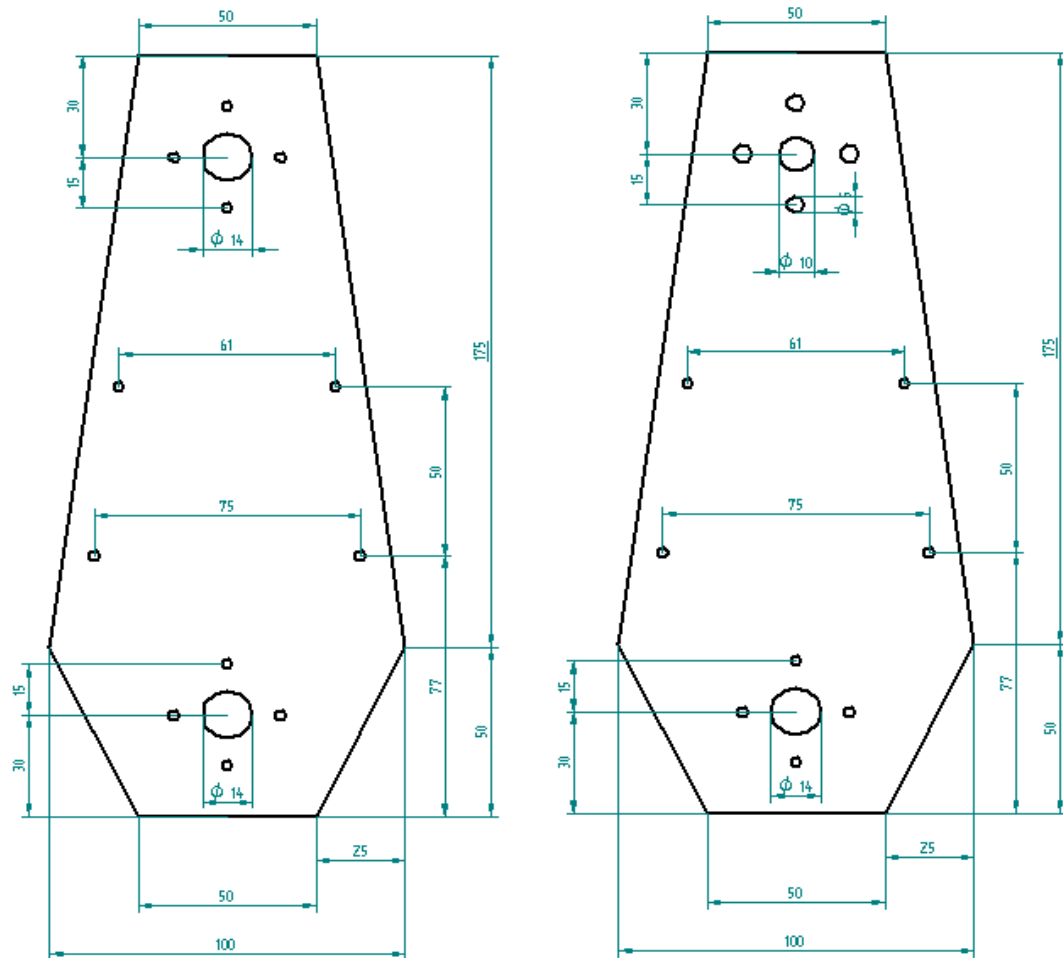
Pieces 'D' & 'E':



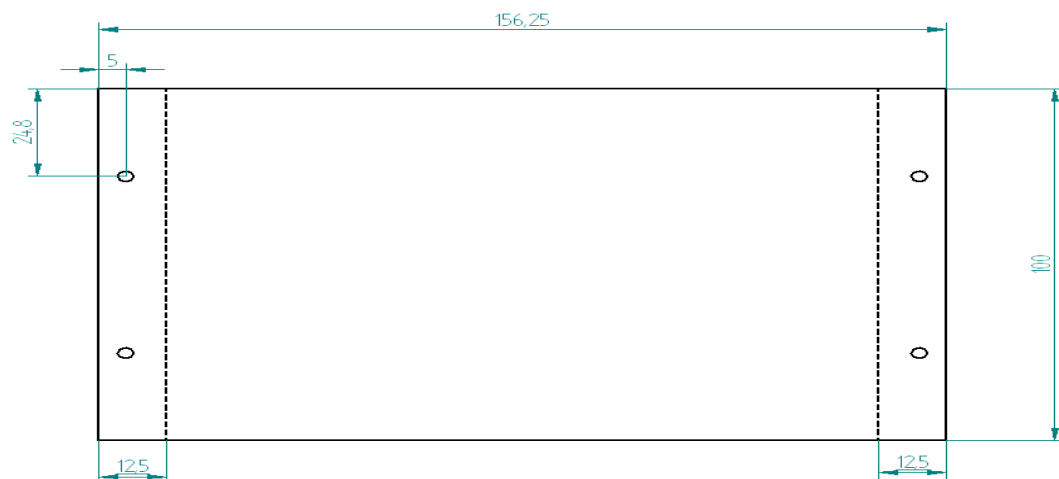
Pieces 'F' & 'G':



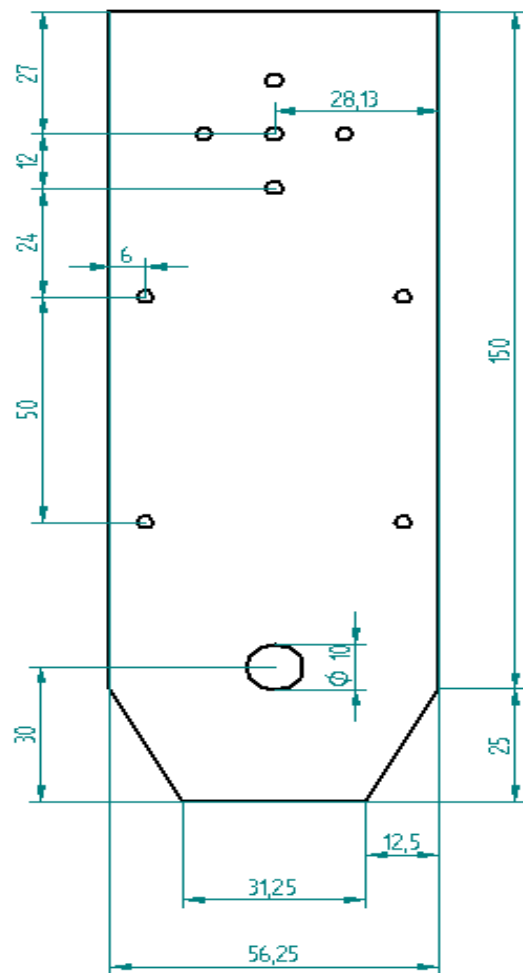
Pieces 'H' & 'I':



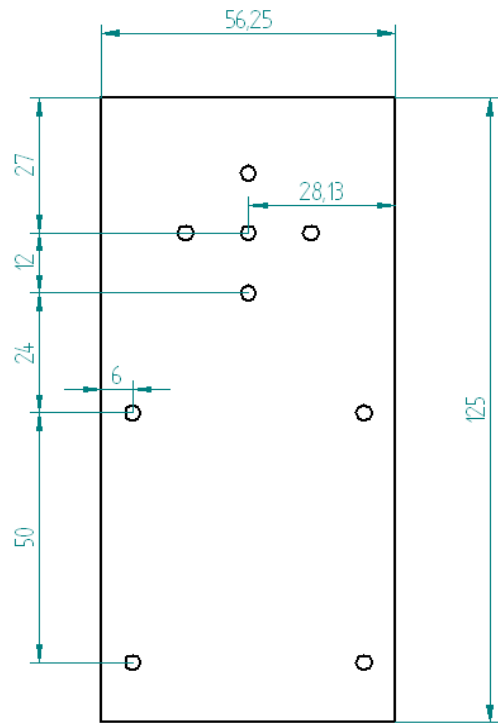
Pieces 'J' & 'K':



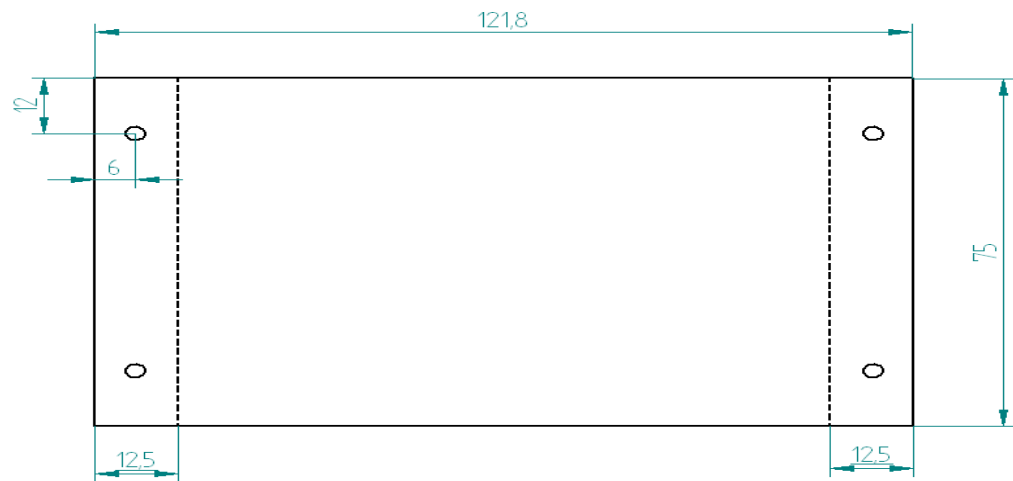
Pieces ‘L’:



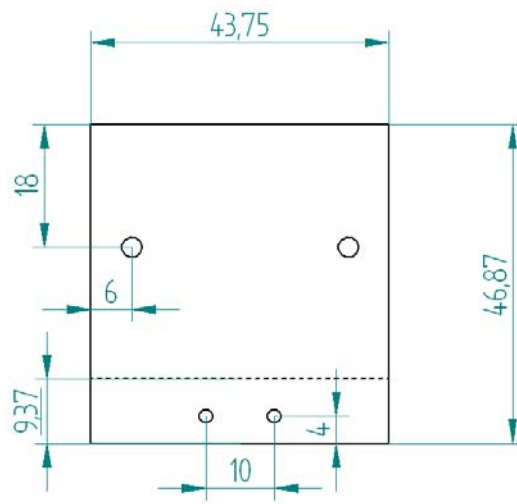
Piece ‘M’:



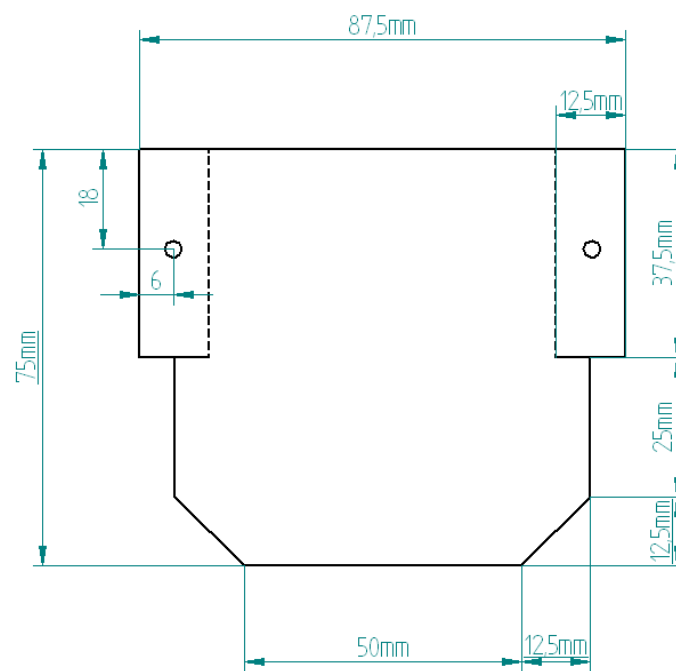
Pieces ‘N’ & ‘O’:



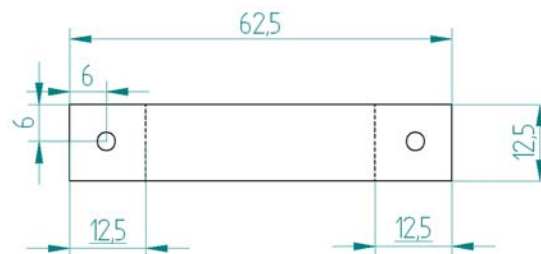
Pieces 'P' & 'Q':



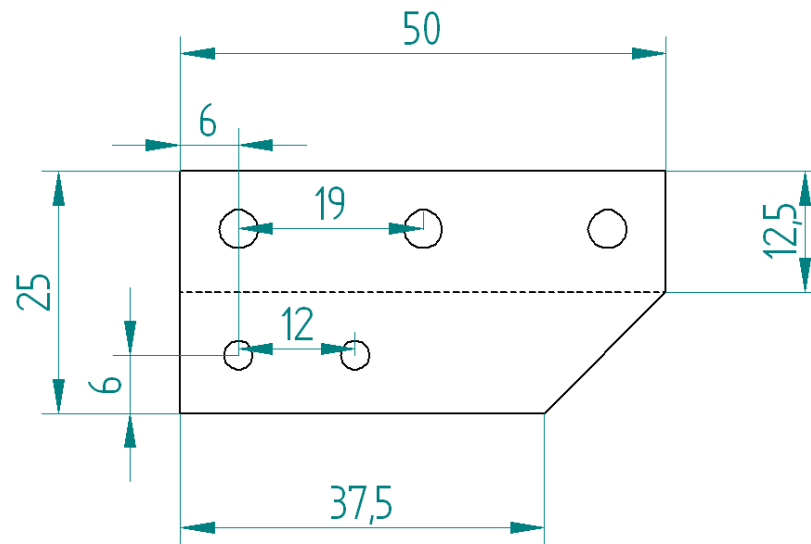
Pieces 'R' & 'S':



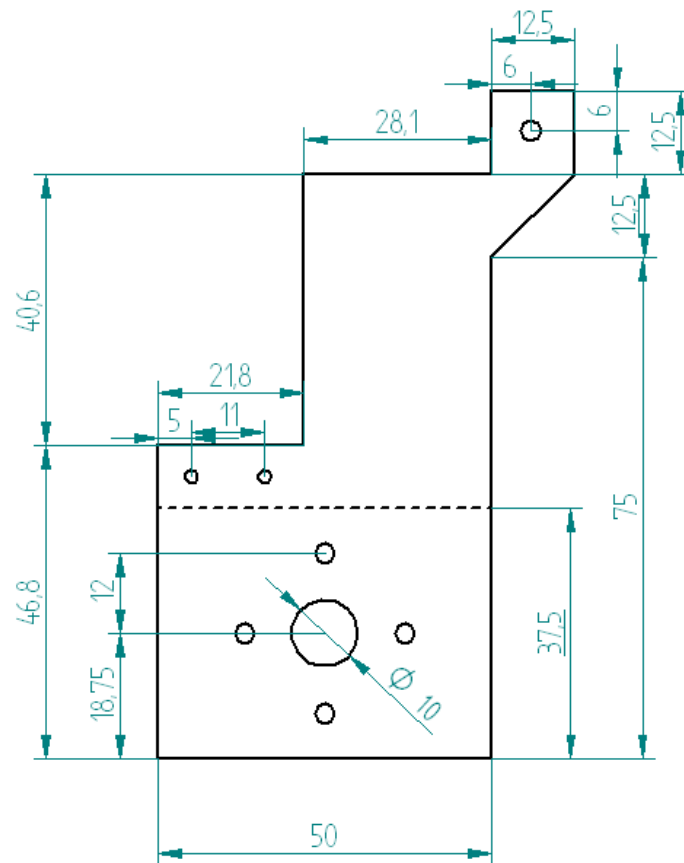
Piece 'T':



Piece 'U', 'V' & 'W':



Piece 'X':



# Appendix E EasyPIC4 Tutorial

EasyPIC4 is a very important development board for the Microchip PIC microcontrollers. It has a lot of features that enables the developers to explore the capabilities of the PIC microcontrollers. The board is shown in Figure E.1.

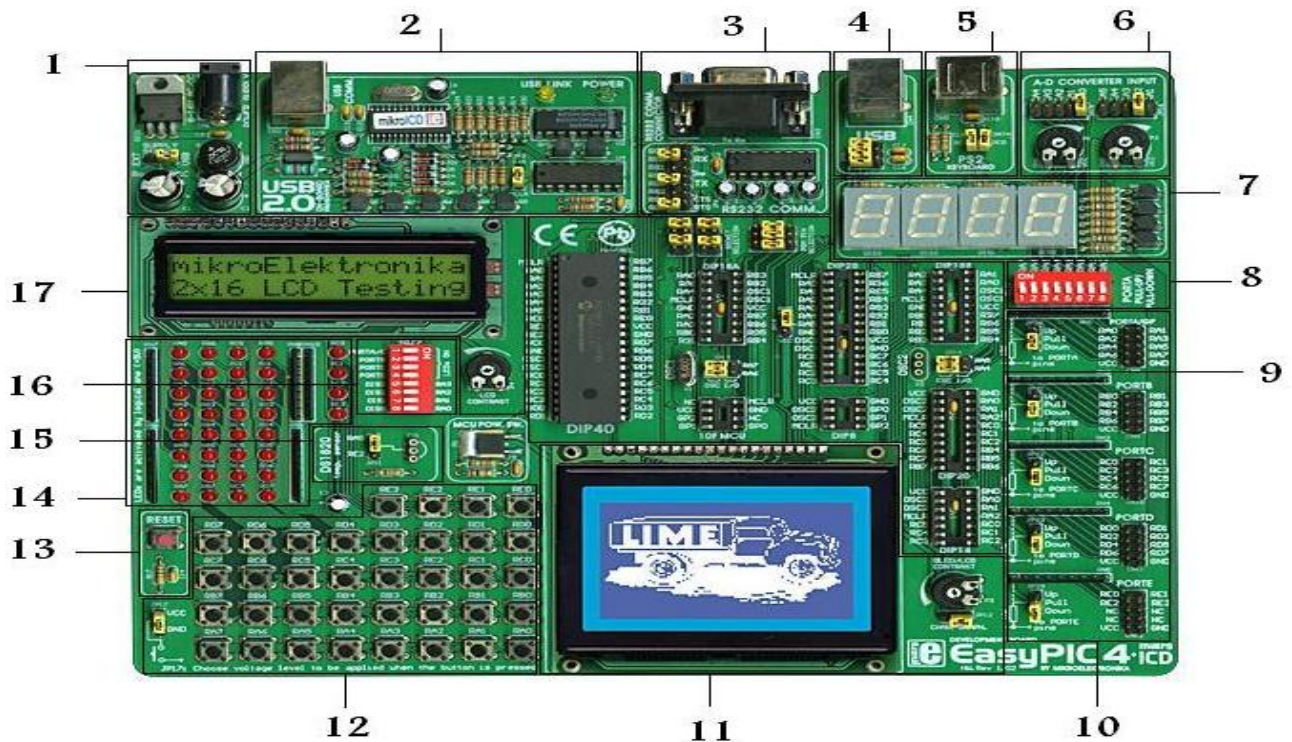


Figure E.1: The EasyPIC4 board

It consists mainly of 17 different circuits. They are:

1. Power supply: The users have the choice from two different power sources, USB or external power supply. The choice is done according to jumper JP1. The USB power is taken from circuit #2. The external power supply may be from 8 to 16V AC/DC. It is regulated to 5V using an LM7805 voltage regulator.
2. USB 2.0 programmer: It is a very fast programmer that has mikroICD (In-Circuit Debugger). The jumper JP2 allows the user to use the MCLR pin as a digital I/O (in that



- case, there is no hardware reset). To load your software into the microcontroller, you have to use the PICflash programming software. The jumper JP5 can be used in case the microcontroller being used is in Low-Voltage programming mode.
3. RS-232 communication: This is a serial female DB9 port used to communicate with the PC. There is MAX232 chip that is used to adjust the voltage levels for the microcontroller. Jumper JP6 can be used for hardware handshaking. The Rx line can be connected to RC7, RB2 or RB1 according to the jumper JP7. The same applies for Tx line which can be connected to Rc6, RB5 or RB2 according to the jumper JP8.
  4. USB communication: This circuit is different than circuit #2 as it is used for communication only while the other is used for programming only. To enable it, just use the jumper JP9 which will connect RC3, RC4 and RC5 pins to the USB.
  5. PS/2 communication: This circuit allows the microcontroller to be connected to PS/2 devices like the keyboard. To enable it, just use the jumper JP10 which will connect RC0 and RC1 to the PS/2.
  6. A/D Converter input: There are 2 potentiometers to be used as an input for the Analog-to-Digital Converter (ADC) of the microcontroller. To use a potentiometer, just connect it to a pin through jumpers JP15 and JP16. JP15 connects the potentiometer P1 to RA0, RA1, RA2, RA3 or RA4. JP16 connects the potentiometer P2 to RA1, RA2, RA3, RA4 or RA5. To disable interference, turn the used pins off on circuit #8.
  7. Seven Segment Display: This is a four 7-segment displays that can be used to display the bits of PORTD. They are enabled through the lower four bits of PORTA and passing through the lower four switches on circuit #16.
  8. PORTA switches: This is a switch used for used for differentiating the connections of PORTA as analog port or as digital port. When a pin is “ON” on this switch, it is connected to pull-up/down resistors (which will affect the analog signal). To use the pin in digital mode, it is recommended to switch it “ON” on this switch to connect it to a pull-up/down resistor (else it will be floating).

9. Microcontrollers' sockets: This circuit has many sockets to fit with any size of microcontroller. There must be only one microcontroller on the board. There are 2 sockets of 18 pins for the different pinouts of the microcontrollers that have 18 pins. There are 2 sockets of 8 pins. One is called "10F MCU" that is used only with PIC10F family. These microcontrollers are not connected to any external crystal as it has only an internal oscillator. The other is called "DIP8" and is used for all other microcontrollers that have 8 pins. There is a fixed crystal of 8MHz that is connected to DIP40, DIP18A, DIP28 and DIP18B. If using one of these microcontrollers, the jumpers JP3 and JP4 must be kept in the default position. An external crystal can be attached to be connected to DIP8, DIP14 and DIP20. If using one of these microcontrollers, the jumpers JP3 and JP4 must be opposite to the default position. Jumpers JP13-14 can be used to make the oscillator pins digital input/output pins. Jumper JP18 is used to connect VCC to RA5 pin for the 18F2331 microcontroller power supply.
10. Direct Port Access: Each pin on the microcontroller can be accessed individually through the connectors placed here. In addition to that, there are VCC and GND connectors beside each port. Jumpers JP20-JP24 are used to connect pull-up/down resistors for each port. This is extremely important for those ports that will be used as inputs. The input port must be connected to the opposite value to the expected input value.
11. Graphical LCD: This is a 128x64 pixel graphical LCD used to display drawings and bitmaps. It is enabled by the jumper JP12 and its contrast can be adjusted using the potentiometer P3. This circuit can also be connected to the LCD 2x16 character in 8-bit mode. The LCD must be placed with 2 free pins to the left and four free pins to the right.
12. Push-Buttons: These are 36 push buttons that are directly connected to every port pin on the microcontroller. They are used to change the state of the corresponding input pin. When pressed, the new state of the pins is controlled via the jumper JP17.
13. Reset Circuit: This is the hardware reset circuit that is connected to the MCLR pin on each microcontroller. It will restart the execution of the code from beginning.

14. LEDs: This circuit is mostly used to display the current status of each port pin. It has 36 led that are directly connected to the microcontroller ports pins. Each port can be directly enabled/disabled using the upper four switches on circuit #16.
15. Temperature sensor: This is a DS1820 Digital Thermometer. It can work in temperature range of -55°C to 125°C with an accuracy of +/-0.5°C. Jumper JP11 is used to direct the output to either RA5 or RE2 pins of the microcontroller.
16. Switch: This switch has two functionalities. The upper four micro-switches are used to control the enabling/disabling of the leds in circuit #14. The lower four micro-switch are used to control the enabling/disabling of the seven segment displays in circuit #7.
17. LCD: This is an LCD of size 2x16 character. Each character is represented by 5x8 pixels. It is connected in 4-bit mode. It can display alphanumeric characters.

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